

Dynamic optimization of agricultural resources within a Water–Energy–Food–Ecosystem Nexus: Uncertainty assessment of carbon and water footprints, and clean energy in the Minjiang River Basin, Western China

Mingkang Yuan ^a, Jiabei Hu ^a, Gavin Kader ^{b,*}, Peng Lei ^c

^a College of Management Science, Chengdu University of Technology, Chengdu, 610059, China

^b Center for Intelligence Economic Science, Southwestern University of Economics and Finance, Chengdu, 611130, China

^c School of Tourism and Cultural Industries, Sichuan Tourism University, Chengdu, 610100, China

ARTICLE INFO

Keywords:

Water-energy-food-ecosystem nexus
Multi-objective optimization
Fuzzy credibility
Sustainable resource management

ABSTRACT

Sustainable agricultural development faces critical challenges from water scarcity, climate change, and poor resource allocation, requiring integrated management under uncertainty. This study proposes an enhanced Water-Energy-Food-Ecosystem Nexus (WEFEN) model incorporating dynamic resource allocation while accounting for uncertainties in resource availability through fuzzy credibility constrained programming. The model solves a dynamic multi-objective optimization program, aiming to maximize economic benefits and clean energy production, while minimizing water and carbon footprints (using NSGA-II). Research findings from the Minjiang River Basin, China, indicate that: (1) optimized resource allocation overall increases agricultural income by 37.61 %, reduces the water and carbon footprints by 17.24 % and 9.20 %, respectively; (2) uncertainty analysis reveals that credibility levels (λ) ranging from 0.70 to 0.80 achieve optimal balance between economic returns and environmental sustainability, while $\lambda > 0.8$ reduces shortage risks but increases external water diversion costs by 6.8 %; (3) dynamic modeling reveals significant seasonal variations in resource allocation efficiency, with water demand fluctuating by up to 60 % across seasons, with large variability in clean energy output (1.21×10^5 kWh in winter with 5.88×10^5 kWh in autumn); (4) except for autumn, concurrent increases in economic benefits and reductions in carbon emissions are achievable, however, autumn's high agricultural productivity creates trade-offs requiring 40 % higher external water diversion to maintain production; (5) regional analysis shows that upstream areas face less environmental pressure due to limited agricultural production, while midstream and downstream areas experience higher water demand and carbon emissions. The study provides policy recommendations, including region-specific water management strategies and optimized seasonal resource allocation promoting sustainability.

1. Introduction

Global population growth, climate change, and resource scarcity collectively intensify the severe challenges facing sustainable development. Sustainable Development Goals (SDGs) such as “Zero Hunger” (SDG₂), “Clean Water and Sanitation” (SDG₆), and “Affordable and Clean Energy” (SDG₇) emphasize ensuring food security, protecting water resources, and efficiently utilizing energy through sustainable practices (UNFCC, 2015). With steady global population growth, by 2030, food demand is projected to increase by 35 %–50 %, energy

demand by 20 %–30 %, and water demand by 40 % (Conway et al., 2015). This unprecedented growth in resource demand not only exacerbates water crises and energy shortages but also severely threatens global food security (Rasul and Sharma, 2016). As food production heavily depends on water, land, energy, and various chemical inputs, this leads to significant increases in anthropogenic carbon emissions (Crippa et al., 2021; Khofi et al., 2025). Reports indicate that agricultural activities account for 34 % of global greenhouse gas emissions, with this proportion growing annually, threatening global carbon neutrality goals (Springmann et al., 2018; Garau et al., 2025). Therefore,

* Corresponding author.

E-mail address: gav.kader@outlook.com (G. Kader).

<https://doi.org/10.1016/j.enpol.2025.114969>

Received 15 May 2025; Received in revised form 22 October 2025; Accepted 8 November 2025

Available online 1 December 2025

0301-4215/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

ensuring food security while achieving sustainable resource use and environmental protection, while simultaneously reducing agricultural carbon emissions and minimizing water footprints, remains a key challenge for sustainable agricultural development (Ortiz-Bobea et al., 2021; Elahi et al., 2024).

Confronting these resource challenges, traditional single-resource management approaches may no longer meet the optimization needs of complex, interdependent systems (Duan et al., 2019). The Water-Energy-Food Nexus (WEFN) analytical framework has emerged to provide a more systematic approach to addressing cross-sectoral resource allocation issues (Hua et al., 2023; Fan et al., 2025). This framework emphasizes the complex interdependencies among water, energy, and food systems, where changes in any single system propagate to others. However, related research indicates that water and energy development often compromises ecosystem integrity (Duan et al., 2020; Guo et al., 2024). Sievers et al. (2025) emphasize that proactive ecosystem management is a prerequisite for maintaining water, energy, and food security. Consequently, research has further incorporated ecosystems, forming the Water-Energy-Food-Ecosystem Nexus (WEFEN) analytical framework, which provides new perspectives for achieving coordinated development between economic growth and environmental protection (Kertolli et al., 2024; Lucca et al., 2023). This development marks an important transition in resource management research from fragmented management toward more integrated management (Ma et al., 2024; Pan et al., 2025). Within this framework, effective cross-sectoral coordination becomes particularly important, as water allocation strategies, energy utilization, and food production must comprehensively consider ecosystem externalities (Martinez-Hernandez et al., 2017; Panjehpour and Zarrinpoor, 2025).

The WEFEN framework exhibits intricate nonlinear linkages among its internal components. Water scarcity directly affects agricultural irrigation capacity and food production scale, while energy consumption is closely linked to water extraction, distribution, and treatment processes (Demir and Alp, 2025). In agricultural production, energy is extensively used for irrigation and mechanized farming, while water availability directly determines the intensity and distribution of energy demand. More complexly, the development and utilization of water and energy resources significantly impact ecosystem functions, while ecosystem service degradation reciprocally affects hydrological cycle stability and agricultural productivity (Napoli et al., 2025; Nika et al., 2022). Climate change further exacerbates the complexity of these interdependencies, disrupting the delicate balance within WEFEN systems and making effective cross-sectoral coordination even more crucial.

WEFEN system research has become an effective approach for modeling the interdependent effects of agricultural, water, energy, and land resources (Ali and Acquaye, 2024; Feng et al., 2025). Earlier research primarily employed conceptual frameworks and theoretical analytical methods, largely focusing on single-factor or dual-factor analyses did not adequately reflect system-wide dynamics and complexity. Simultaneously, research on interdependencies among subsystems within the WEFEN framework remains relatively limited, particularly in term of sustainable ecological management. Notably, while existing research increasingly incorporates carbon and water footprints into WEFEN analyses, papers predominantly employ separated analytical approaches when addressing these two types of footprints, overlooking synergistic opportunities for joint optimization (Garcia and You, 2016; Zolghadr-Asli et al., 2023; Wu et al., 2025). To better achieve regional sustainable development goals and construct low-carbon, environmentally friendly WEFEN systems, integrating carbon and water footprints into the WEFEN analytical framework has become a critical issue requiring urgent resolution, which constitutes one of the core focuses of this study.

When optimizing agricultural WEFEN system resource allocation, it is necessary to incorporate low-carbon agricultural development goals and sustainability oriented toward ecological environmental protection to ensure resources are properly utilized in sustainable and

environmentally friendly ways. Correa-Cano et al. (2022) incorporated ecosystem health constraints when constructing a WEFE framework, specifically requiring that ecological water volumes in rivers and lakes should not fall below minimum ecological water requirements. Qin et al. (2022) combined water pollution treatment costs (ecological compensation fees) with system economic benefits as objective functions, providing references for water and land resource allocation strategies that consider water resource pollution. However, existing research on interdependencies within the WEFE framework is still developing, with limited applied research on sustainable ecological environmental management. Melo et al. (2021) investigated interconnections between WEFN and specific ecosystems, exploring technical strategies and advancing methodological frameworks to achieve balance among goals such as economic growth and environmental protection. Yang et al. (2020) proposed multi-objective optimization methods based on fuzzy credibility programming, addressing uncertainty factors and enhancing system adaptability and flexibility under climate change contexts. Similarly, Cui et al. (2023) incorporated water footprints into analytical frameworks and proposed agricultural planting structure optimization models to improve comprehensive resource utilization efficiency and promote sustainable management of food production and water resources. However, current research predominantly considers water footprints or carbon footprints separately, implicitly excluding possibilities for studying joint optimization mechanisms (Yoon et al., 2022). He et al. (2022) provides the first systematic assessment of how crop planting patterns impact multiple systems from a Food-Energy-Water-Carbon nexus perspective, revealing critical trade-offs where cropping structure optimization can improve economic benefits but simultaneously reduce carbon sequestration capacity. Their work demonstrates that cropping structure changes generate complex cascading effects at the system level, revealing that nexus frameworks have strong explanatory capacity for identifying multi-objective conflicts. Specifically, He et al. (2022) focuses on assessing what to plant and where (spatial crop distribution), thereby providing policy implications based on impact assessments of how different crop structures affect economic benefits and carbon emissions. In contrast, our study addresses the complementary question of how to manage resources within existing agricultural systems through dynamic optimization strategies, thereby providing policy implications based on operational resource allocation decisions rather than spatial crop structure adjustments. Overall, the above literature highlights the need for comprehensive ecosystem resource allocation research that includes energy and land.

In recent years, integrated modeling and multi-objective optimization methods have emerged, with system dynamics models, input-output analysis, and life cycle assessment methods being widely applied to WEFEN system analyses. These methods have underpinned analyses that address multi-dimensional conflicts and synergistic effects in WEFEN systems (Sušnik, 2018; Vahedi et al., 2025). In WEFEN system optimization, the introduction of mathematical methods such as dynamic programming and stochastic programming provides new approaches for solving complex resource allocation problems. System dynamics models are widely applied to simulate dynamic changes in WEFE systems, capable of evaluating long-term impacts of policy interventions across different time scales (Wichelns, 2017). Scenario analysis is commonly used to assess performance among the water, energy, food, and environmental domains under different policy scenarios to inform sustainable resource allocation strategies (Zhang et al., 2020). However, existing optimization methods are predominantly based on static programming, failing to adequately consider dynamic impacts of seasonal variations and climate uncertainty on resource allocation. Assuming (implicitly or explicitly) that the system is static is inadequate amid increasingly frequent extreme climate events and pronounced seasonal resource fluctuations, which undermines any policy design by precluding forward-looking and adaptive resource allocation. Therefore, how to effectively integrate dynamic characteristics and uncertainty factors

into WEFEN optimization frameworks is one of the focal points of this research.

Multi-objective optimization theory provides a rigorous toolkit for modeling sustainable management of WEFEN systems (Mansour et al., 2023). By constructing multi-objective functions containing economic, environmental, and social benefits, researchers can optimize water resource allocation under carbon footprint constraints, and incorporate water footprints into agricultural planting structure optimization, achieving synergistic development of resource optimization. Regarding solution algorithms, traditional single-objective optimization algorithms no longer meet the complex requirements of WEFEN systems, prompting researchers to develop more advanced multi-objective evolutionary algorithms (e.g. NSGA-II). These algorithms can maintain solution diversity when handling multiple conflicting objectives, providing decision-makers with rich Pareto optimal solution sets. However, existing algorithms still face challenges such as slow convergence speeds and high computational complexity when handling high-dimensional multi-objective problems. Particularly in dynamic multi-stage optimization processes, how to balance algorithmic exploration and exploitation capabilities to achieve effective trade-offs among multiple objectives represents another focal point of this research.

Table 1 provides a summary of comparisons with other studies; this highlights that while the literature has made progress across a wide-range of problems, our paper provides a much more comprehensive approach to existing WEFEN research. Based on this, this study addresses limitations in existing WEFEN research by proposing a novel dynamic multi-objective optimization framework and validates the effectiveness of the proposed method using the Minjiang River Basin in western China as a case study, aiming to provide scientific evidence and policy recommendations for sustainable resource management of basin agricultural WEFEN systems. The specific contributions and innovation are as follows:

(1) Simultaneously incorporating carbon and water footprints as independent environmental objectives in WEFEN systems, rather than treating them as secondary constraints, achieving precise trade-offs between economic returns and environmental impacts; (2) Introducing dynamic programming techniques to explicitly model seasonal variations and temporal transition processes, which can more accurately reflect dynamic evolutionary characteristics of time-dependent decision-making in the real world compared to traditional static programming methods; (3) Employing fuzzy credibility programming to jointly consider climate variability, stochastic water inflows, and seasonal uncertainties, providing an effective framework for robust optimization under resource scarcity risks.

This paper is organized as follows: Section 2 introduces the research methodology and model framework. Section 3 explains the complete solution method using the NSGA-II. Section 4 applies the WEFEN model to the Minjiang River Basin, China, as a case study. Section 5 discusses the significance of the results. Section 6 summarizes the main conclusions of this paper and provides policy recommendations.

2. Research framework and model construction

2.1. Research framework

This study proposes a dynamic multi-objective optimization model based on the WEFEN, which extends the traditional WEFN to better coordinate between economic benefits, energy utilization, water resource allocation, and carbon emission control (Ren et al., 2022). We incorporate carbon and water footprints objectives within the WEFEN system to create a more realistic management framework that reinforces sustainable development. Specifically, the objectives are to maximize economic benefits in irrigation areas, maximize renewable energy production, minimize water footprint, and minimize carbon footprint (Bieber et al., 2018). The decision variables are the water resource allocation amounts and cultivated land allocation amounts for different time periods, different regions, and different sectors.

To maintain computational tractability and align with basin management capabilities, this study delineates clear model boundaries and assumptions. The carbon footprint assessment focuses exclusively on direct agricultural emissions from fertilizer application, pesticide use, and diesel consumption within the basin boundary. While comprehensive lifecycle analysis would ordinarily include embedded emissions from irrigation infrastructure manufacturing, seed production, and transportation systems, for the purpose of our model, we assume that such emissions are negligible relative to direct emissions. This reflects three key considerations that justify our model assumption: (1) controllability for basin managers: emphasis is placed on operational levers (fertilizer, pesticides, irrigation, and fuel use) whose emissions are directly shaped by local and basin-level decisions, ensuring that the modeled sources are those most responsive to agricultural management, allowing us to provide impactful policy recommendations; (2) data availability and methodological consistency: direct agricultural inputs have widely used, standardized emission factors (IPCC guidelines), supporting reproducible, basin-level estimates across seasons and regions. However, the typically detailed life-cycle data that would be required when including indirect emissions would vary substantially by technology, geography, and supplier, making them hard to standardize for basin-wide, seasonal comparisons, not ignoring the fact that such data are not readily observable; (3) optimization: the model's decision space centers on seasonal and spatial resource allocation and associated operational inputs. However, Lifecycle emissions are largely independent of those seasonal and spatial resource allocation decisions. Incorporating them would add dimensionality without materially changing the Pareto frontier or the policy trade-offs in a substantive way, thus we preserve tractability and whilst still allowing us to provide realistic policy suggestions.

Given the complexity and temporal uncertainty of agricultural water systems, this model incorporates the following elements:

Table 1
Comparative analysis of literature.

Key literatures	Multi-objective	Ecological Environment	Carbon Footprint	Water Footprint	Dynamic Programming	Uncertainty Analysis	Seasonal Variation	Credibility constraint
Hua et al. (2022); Zhang et al. (2025)	✓	✓	✓			✓		✓
Correa-Cano et al. (2022); Fan et al. (2018)	✓	✓		✓			✓	✓
Qin et al. (2022); Cui et al. (2023)	✓	✓		✓		✓		
He et al. (2022); Sanchez-Zarco and Ponce-Ortega (2023)	✓		✓	✓	✓	✓	✓	✓
Cui and Wang (2024); Cao et al. (2024)		✓		✓				✓
Yuan et al. (2023); Tan et al. (2025)	✓	✓			✓	✓	✓	
Yang et al. (2025); Shannak (2022);	✓	✓		✓		✓		
This study	✓	✓	✓	✓	✓	✓	✓	✓

- (1) A stochastic expected value model to address the random utilization rate of water resources in the basin;
- (2) Expression of uncertainty due to agricultural seasonal changes through dynamic programming;
- (3) Uncertain inputs characterized by triangular fuzzy numbers;
- (4) Introduction of Fuzzy Credibility Constrained Programming (FCCP) to handle the possibility and necessity of uncertain events;
- (5) Employment of 0–1 integer programming to indicate whether external water resources need to be utilized.

The proposed WEFEN framework is applied to the irrigation area of the Minjiang River Basin which verifies its practical applicability. The specific research framework is shown in Fig. 1:

2.2. Research methods

2.2.1. Dynamic programming

We use dynamic programming techniques to characterize the uncertainty of seasonal changes in agricultural activities (Yuan et al., 2023; Bertsekas, 1982). Seasonal changes in agricultural production activities allow for a more accurate characterization and optimization by incorporating the dynamic links explicitly within the WEFEN.

2.2.2. Multi-objective optimization model

Multi-objective optimization models which require the optimization of more than one objective function (Mansour et al., 2023), are used to balance interactions among socio-economic, resource, and environmental factors (Yang et al., 2025), aiming to achieve maximum economic benefits, maximum renewable energy output, minimum water footprint, and minimum carbon footprint.

2.2.3. Fuzzy credibility constrained programming

Fuzzy Credibility Constrained Programming (FCCP) is a mathematical programming approach used when data in system models reflect uncertainty where precise data probability distributions cannot be obtained appropriately. Fuzzy theory typically includes three types of measures for handling uncertainty: possibility, necessity, and credibility. Possibility and necessity have limitations in terms of lack of self-duality; they are unable to equally reflect optimistic or pessimistic perspectives of uncertainty. In contrast, credibility integrates both possibility and necessity, providing a more balanced and effective approach for measuring uncertainty in fuzzy problems. According to the concept of credibility (Yang et al., 2020), the probability of fuzzy events can be represented as follows:

$$\begin{aligned} & \max \sum_{j=1}^n c_j x_j \\ & Cr \left\{ \sum_{j=1}^n \tilde{a}_{ij} x_j \leq b_i \right\} \geq \lambda_i, (i = 1, \dots, m) \\ & x_j \geq 0, (i = 1, \dots, m) \end{aligned} \quad (1)$$

where $x_j = (x_1, x_2, \dots, x_n)$ is the non-fuzzy decision variable vector and c_j are weights in the objective function. Cr is the credibility measure with coefficients $\tilde{a}_{ij}$, b_i for each of the constraints, with λ being the confidence/credibility level, quantifying the uncertainty i.e. when λ is higher/lower there is lower/higher tolerance for uncertainty.

2.2.4. 0–1 integer programming

0–1 Integer Programming (0–1 IP) is a mathematical optimization method used when decision variables are binary, i.e., either 0 or 1. This technique is commonly applied in situations where decisions are yes/no in nature, such as in resource allocation or task assignment problems (Yusuf et al., 2024). 0–1 IP is used in problems such as knapsack, set cover, and scheduling, where decisions are discrete and involve

choosing between two options.

The general formulation is:

$$\max Z = \sum_{j=1}^n c_j x_j \quad (2)$$

Subject to:

$$\begin{aligned} & \sum_{j=1}^n a_{ij} x_j \leq b_i \quad \forall i \in \{1, 2, \dots, m\} \\ & x_j \in \{0, 1\} \quad \forall j \in \{1, 2, \dots, n\} \end{aligned} \quad (3)$$

where c_j are weights for decision variables x_j , and a_{ij} are the coefficients on each x_j for each of the constraints.

2.3. WEFEN dynamic optimization model

2.3.1. Symbol description

Table 2 provides a description of all the notation used in the model. Subscripts $t/n/i$ refer to time/region/non-agricultural sector; the time dimensions comprises the four seasons of the year (spring, summer, autumn, winter), the spatial dimensions cover 8 cities in the MRB (Chengdu, Zigong, Neijiang, Leshan, Meishan, Yibin, Ya'an, Aba), and the non-agricultural sector includes the domestic and industrial sectors that are not part of agriculture.

2.3.2. Objective functions

(1) Economic Benefits for Irrigation Areas

Economic benefits are an important indicator reflecting economic development defined as the difference between income (across all regions, all sectors, and time) and associated expected costs of production. Income is obtained from crop production (R_c) and non-agricultural sector production (R_s). Costs arise from planting, irrigation, and transportation. θ_m is a 0–1 decision variable used to determine whether external water input is needed in region n at time t . Maximizing economic benefits is expressed as:

$$\begin{aligned} \max F_1 &= \max(R_c + R_s - C_p - C_e - C_w) \\ R_c &= \sum_{n=1}^N \left(\sum_{t=1}^T M_m \right) \cdot EY_n \cdot y_0 \\ R_s &= \sum_{t=1}^T \sum_{n=1}^N \sum_{i=1}^I NA_{mi} \cdot b_0 + \sum_{t=1}^T \sum_{n=1}^N WA_m \cdot w_0 \\ C_p &= \sum_{t=1}^T \sum_{n=1}^N M_m \cdot c_p \\ C_e &= \sum_{t=1}^T \sum_{n=1}^N M_m \cdot c_w \left[\frac{SW_m}{\alpha_1} + \frac{IW_m}{\alpha_2} \right] \\ C_w &= \sum_{t=1}^T \sum_{n=1}^N M_m \cdot c_d \cdot \frac{BW_m}{\alpha_1} \cdot \theta_m \\ WA_m &= SW_m + BW_m \end{aligned} \quad (4)$$

(2) Clean Energy Production

The trade-offs between the non-renewable energy market and global climate change means finding ways to generate clean energy are becoming more popular among policymakers. The objective function below considers two typical types of clean energy: hydropower (generated from water) and biomass (generated from straw). The formula for maximizing clean energy production is:

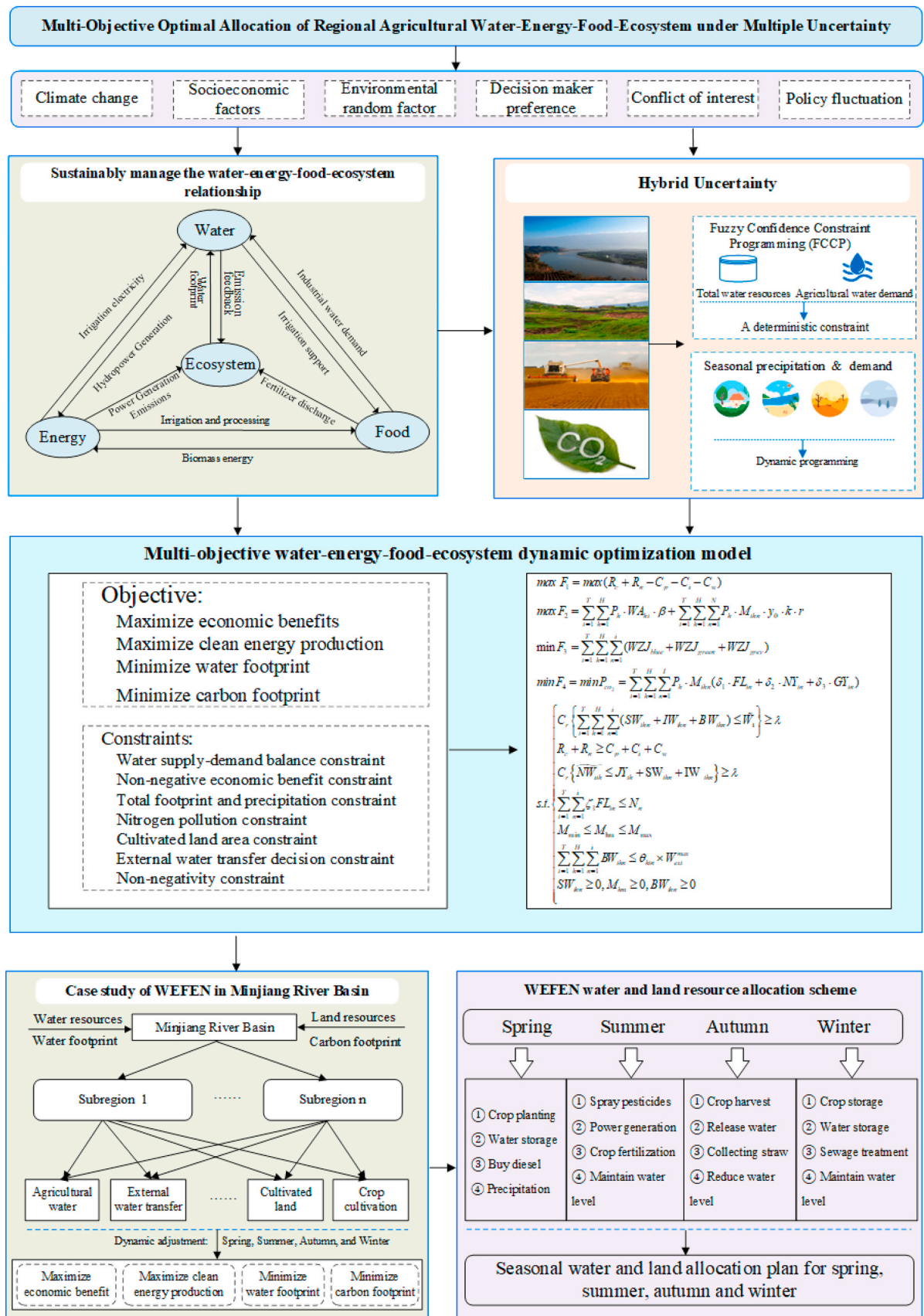


Fig. 1. WEFEN system coupling framework.

Table 2
Model notation and variable definition.

Decision variable	Definition
M_n	Irrigated area n in period t for region (ha)
SW_n	Water quantities allocated from reservoirs (m^3)
BW_n	External water transfer in irrigated areas (m^3)
θ_n	0-1 variable determining external water input requirement
Symbol	Definition
R_c	Crop production income (CNY)
R_s	Non-agricultural sector production income (CNY)
C_p	Cost of planting crops (CNY)
C_e	Irrigation expenses in irrigated areas (CNY)
C_w	External water diversion costs (CNY)
W_n	Total water resources in period t for region n
NA_{ni}	Water allocation for non-agricultural sector i (m^3)
WA_n	Hydropower stations water resources of in period t for region. (m^3)
IW_n	Water storage in period t for region n (m^3)
WZJ_{blue}	Blue water footprint (m^3)
WZJ_{green}	Green water footprint (m^3)
WZJ_{grey}	Grey water footprint (m^3)
NW_n	Minimum water requirement for crops in period t for region n . (m^3)
JY_n	Natural precipitation in period t for region n . (m^3)
FL_n	Fertilizer usage in period t for region n . (kg)
NY_n	Pesticide usage in period t for region n . (kg)
GY_n	Diesel usage in period t for region n . (L)
Model parameters	Definition
EY_n	Crop market price in region n (CNY/ton)
y_0	Crop yield per unit area (Ton/ha)
b_0	Non-agricultural sector economic benefit coefficient (CNY/ m^3)
w_0	Hydropower economic benefit coefficient (CNY/ m^3)
c_p	Cost of planting crops per unit area (CNY/ha)
c_w	Irrigation cost per unit area (CNY/ha)
α_1	Water utilization coefficient 1 (%)
α_2	Water utilization coefficient 2 (%)
c_d	External water diversion cost per unit area (CNY/ m^3)
β	Power generation coefficient of hydropower station (kWh/ m^3)
k	Crop straw particle ratio (%)
r	Calorific value of crops burning bioenergy (kWh/kg)
ζ_1	Nitrogen leaching coefficient of nitrogen-containing complex (kg CE/kg)
N_n	Suitable nitrate concentration for crops (%)
δ_1	Carbon emission factor of fertilizer (kg CE/kg)
δ_2	Carbon emission factor of pesticides (kg CE/kg)
δ_3	Carbon emission factor of diesel (kg CE/L)

$$\max F_2 = \sum_{t=1}^T \sum_{n=1}^N WA_n \cdot \beta + \sum_{t=1}^T \sum_{n=1}^N M_n \cdot y_0 \cdot k \cdot r \quad (5)$$

where β is the power generation coefficient of hydropower stations, k is the straw particle ratio of crops, and r is the bioenergy calorific value when burning agricultural crops. The coefficient β represents the net water-to-energy conversion efficiency (kWh/ m^3) for hydropower generation with this parameter implicitly incorporating water-saving effects from hydropower operations.

(3) Total Crop Water Footprint

Crop water footprint refers to the sum of direct and indirect water consumption throughout the entire crop growth process. It is a comprehensive indicator for quantifying freshwater consumption and water pollution. Crop water footprint typically consists of three parts: green water footprint, blue water footprint, and grey water footprint. Green water footprint refers to the consumption of green water resources (e.g., rainfall absorbed by crops); blue water footprint is the consumption of blue water resources (e.g., surface water or groundwater used for irrigation); grey water footprint is the amount of freshwater required to dilute pollutant loads (e.g., nitrogen compounds) based on natural background concentrations and existing environmental water quality standards. To achieve sustainable use of water resources and promote protection of the environment, minimizing the water footprint

per unit crop yield is set as the third objective function:

$$\begin{aligned} \min F_3 &= WZJ_{blue} + WZJ_{green} + WZJ_{grey} \\ WZJ_{blue} &= \sum_{t=1}^T \sum_{n=1}^N NW_n - JY_n \\ WZJ_{green} &= \sum_{t=1}^T \sum_{n=1}^N \min(NW_n, JY_n) \\ WZJ_{grey} &= \sum_{t=1}^T \sum_{n=1}^N \frac{\zeta_1 FL_n}{N_n} \end{aligned} \quad (6)$$

Where WZJ_{blue} , WZJ_{green} , and WZJ_{grey} are the blue water footprint, green water footprint, and grey water footprint, respectively, NW_n is the crop water requirement, JY_n is the natural precipitation for region n during period t , ζ_1 is the nitrogen leaching coefficient of nitrogen compounds, FL_n is the amount of fertilizer used in region n during period t , and N_n is the appropriate nitrate concentration for crops in region n .

(4) Total Carbon Footprint

Crop carbon footprint is an important indicator for guiding sustainable farmland management, which is defined as estimated carbon emissions caused by production practices during the crop lifecycle, with total carbon emissions mainly due to agricultural inputs or activities, including fertilization, pesticide application, diesel for agricultural

machinery, and electricity for irrigation. Carbon sequestration can be estimated by multiplying emission factors per unit area by respective rice planting areas. Thus, the fourth objective is to minimize the carbon footprint per unit crop yield:

$$\min F_4 = \min P_{co_2} = \sum_{t=1}^T \sum_{n=1}^N M_n (\delta_1 \cdot FL_n + \delta_2 \cdot NY_n + \delta_3 \cdot GY_n) \quad (7)$$

where δ_1 is the coefficient of CO₂ production from fertilizer utilization, δ_2 is the coefficient of CO₂ production from pesticide utilization, δ_3 is the coefficient of CO₂ production from diesel and gasoline used in agricultural activities, FL_n is the fertilizer use in region n during period t , NY_n is the pesticide use in region n during period t , and GY_n is the diesel and gasoline use in the region during period t .

2.3.3. Constraints

- (1) **Water Resource Supply and Demand Balance Constraint:** The allocation of water resources must satisfy the total water resource supply in the irrigation area. This constraint uses FCCP, which effectively reveals how the optimal water allocation scheme varies with changes in total water resources. The FCCP fuzzy credibility programming is used to represent the uncertainty in the total water resources of the region.

$$C_r \{SW_n + IW_n + BW_n \leq \tilde{W}_n\} \geq \lambda \quad (8)$$

where C_r is the credibility measure, λ is the confidence level, $\tilde{W}_n$ is the total available water in the irrigation area, and the total water allocation for all regions cannot exceed the total water supply.

- (2) **Non-negative Economic Returns Constraint:** The total income from agricultural and non-agricultural production must at least cover all costs to avoid losses. In other words, we assume that total production generates sufficient revenues to at least break even.

$$R_c + R_s \geq C_p + C_e + C_w \quad (9)$$

- (3) **Total Water Footprint and Precipitation Constraint:** Ensures that crop water requirements can be met through precipitation and irrigation. The FCCP fuzzy credibility programming is used to represent the uncertainty in the minimum agricultural water requirement for the region.

$$C_r \{\tilde{N}\tilde{W}_n \leq JY_n + IW_n\} \geq \lambda \quad (10)$$

Where $\tilde{N}\tilde{W}_n$ is the minimum agricultural water requirement for the region.

- (4) **Nitrogen Pollution Constraint:** Fertilizer use cannot lead to excessive nitrate concentrations.

$$\sum_{t=1}^T \zeta_1 FL_n \leq N_n \quad (11)$$

where N_n is the nitrate concentration standard for the region, controlling agricultural pollution.

- (5) **Cultivated Land Area Constraint:** The cultivable area in the region is bounded from above and below, based on the geographical area.

$$M_{\min} \leq M_n \leq M_{\max} \quad (12)$$

where $M_{\min}$ and $M_{\max}$ are the minimum and maximum cultivable areas for the region, respectively.

- (6) **External Water Diversion Decision Constraint:** The irrigation area's need for external water diversion must satisfy the 0–1 variable constraint meaning the external water diversion volume should not exceed the maximum available external water volume.

$$\sum_{t=1}^T \sum_{n=1}^N BW_n \leq \theta_n \times W_{\text{ext}}^{\max} \quad (13)$$

where $W_{\text{ext}}^{\max}$ is the maximum external water volume that can be diverted.

- (7) **Non-negativity Constraints:** Decision variables are non-negative.

$$SW_n \geq 0, M_n \geq 0, BW_n \geq 0 \quad (14)$$

2.3.4. State transition equation

As we are concerned with the dynamics of resource allocation, we define a state transition equation as:

$$\tilde{W}_{t+1,n} = \tilde{W}_n + SW_n + IW_n + BW_n \quad (15)$$

where $\tilde{W}_{t+1,n}$ represents the available water in region n in the next period, and $\tilde{W}_n$ is the available water in region n in the current period t .

We model in this way as the remaining water from the previous period can be transferred to the current period because the water resource allocation process is a continuous dynamic process (Liu et al., 2022). Dynamic programming captures the water resource allocation process effectively, as the state variable represents the current available water in the region, with the state transition equation also accounting for remaining water from the previous period.

2.3.5. Global model

By integrating the aforementioned objectives and constraints, we develop a comprehensive dynamic optimization model within the WEFEN framework that coordinates the complex interdependencies among water resources, energy, food production, and ecological protection.

The global model is shown in equation (16). In terms of model objectives, economic benefit maximization encompasses maximizing agricultural planting income (R_c and R_n), minimizing agricultural costs and external water diversion costs (C_p , C_i and C_w); clean energy maximization mainly considers hydroelectric power generation and biomass energy output; water footprint minimization balances green water, blue water, and grey water footprints to reduce excessive water resource consumption; carbon footprint minimization is represented by carbon emissions in the agricultural production process.

$$\begin{aligned}
\max F_1 &= \max(R_c + R_s - C_p - C_e - C_w) \\
\max F_2 &= \sum_{t=1}^T \sum_{n=1}^N WA_{tn} \cdot \beta + \sum_{t=1}^T \sum_{n=1}^N M_{tn} \cdot y_0 \cdot k \cdot r \\
\min F_3 &= WZJ_{blue} + WZJ_{green} + WZJ_{grey} \\
\min F_4 &= \sum_{t=1}^T \sum_{n=1}^N M_{tn} (\delta_1 \cdot FL_{tn} + \delta_2 \cdot NY_{tn} + \delta_3 \cdot GY_{tn}) \\
\text{s.t.} \left\{ \begin{aligned}
R_c &= \sum_{n=1}^N \left(\sum_{t=1}^T M_{tn} \right) \cdot EY_n \cdot y_0 \\
R_s &= \sum_{t=1}^T \sum_{n=1}^N \sum_{i=1}^I NA_{tni} \cdot b_0 + \sum_{t=1}^T \sum_{n=1}^N WA_{tn} \cdot w_0 \\
C_p &= \sum_{t=1}^T \sum_{n=1}^N M_{tn} \cdot C_p \\
C_e &= \sum_{t=1}^T \sum_{n=1}^N M_{tn} \cdot C_w \left[\frac{SW_{tn}}{\alpha_1} + \frac{IW_{tn}}{\alpha_2} \right] \\
C_w &= \sum_{t=1}^T \sum_{n=1}^N M_{tn} \cdot C_d \cdot \frac{BW_{tn}}{\alpha_1} \cdot \theta_n \\
WA_{tn} &= SW_{tn} + BW_{tn} \\
WZJ_{blue} &= \sum_{t=1}^T \sum_{n=1}^N NW_{tn} - JY_{tn} \\
WZJ_{green} &= \sum_{t=1}^T \sum_{n=1}^N \min(NW_{tn}, JY_{tn}) \\
WZJ_{grey} &= \sum_{t=1}^T \sum_{n=1}^N \frac{\zeta_1 FL_{tn}}{N_n} \\
\tilde{W}_{t+1,n} &= \tilde{W}_{tn} + SW_{tn} + IW_{tn} + BW_{tn} \\
C_r \{SW_{tn} + IW_{tn} + BW_{tn} \leq \tilde{W}_{tn}\} &\geq \lambda \\
R_c + R_s &\geq C_p + C_e + C_w \\
C_r \{NW_{tn} \leq JY_{tn} + IW_{tn}\} &\geq \lambda \\
\sum_{t=1}^T \zeta_1 FL_{tn} &\leq N_n \\
M_{\min} &\leq M_{tn} \leq M_{\max} \\
\sum_{t=1}^T \sum_{n=1}^N BW_{tn} &\leq \theta_n \times W_{\max}^{ext} \\
SW_{tn} &\geq 0, M_{tn} \geq 0 \\
BW_{tn} &\geq 0
\end{aligned} \right. \quad (16)
\end{aligned}$$

Our model integrates dynamic multi-temporal-spatial constraint optimization i.e. joint optimization spanning multiple time periods (T) and regions (N). Through effective constraint management of different water sources, the model prevents water resource overload and waste while incorporating carbon footprint concerns to reduce greenhouse gas emissions, achieving environmental sustainability while meeting production demands. Additionally, to address uncertainties in external environment and internal demand changes, the model employs dynamic programming and credibility constraint methods, utilizing uncertainty constraints to ensure solution robustness, effectively preventing solution deviations caused by uncontrollable factors.

Our dynamic multi-objective optimization framework does not

explicitly model water evaporation and water-saving effects from hydropower generation as separate decision variables. This methodological choice reflects several practical and theoretical considerations. First, our water availability parameters ($\tilde{W}_{tn}$ in Equation (8)) is based on net available water resources that inherently reflect all system losses, including reservoir evaporation, as well as the net outcomes of existing hydropower operations. Therefore, our optimization allocates water that is already net of evaporation losses and water savings from hydropower stations. Second, reservoir evaporation represents a passive system loss that occurs regardless of operational decisions and does not directly influence the optimization variables (SW_{tn} , BW_{tn}) that drive resource allocation across sectors. Third, our fuzzy credibility-constrained programming framework explicitly captures residual variability in net water availability from multiple sources, inclusive of evaporation fluctuations, and hydropower water-saving effects. While explicit modeling of evaporation dynamics could theoretically provide additional refinement, this would require disaggregated reservoir-level data that are not readily available and would substantially increase model complexity without materially altering the agricultural resource allocation decisions that constitute our primary focus.

Since the model contains four counterbalancing objective functions and multiple complex constraints, it exhibits significant non-linear, multi-variable coupling characteristics with an extensive optimization space, making it difficult to solve. Furthermore, the multi-dimensional non-linear and non-convex features in the model lead to a vast number of variables and constraints making the optimization problem difficult to solve through convex programming methods (Chen et al., 2023). Conventional linear programming or mixed-integer programming methods struggle to effectively solve such non-linear multi-objective optimization problems; therefore, evolutionary algorithms (such as NSGA-II, MOEA/D) or meta-heuristic algorithms (such as simulated annealing, particle swarm optimization) are typically employed, combined with parallel computing techniques to accelerate algorithm convergence and enhance solution efficiency.

3. Solution method

The main issues in solving the above model lie in i) addressing the uncertainty and, ii) the multi-objective Pareto optimality. This study employs a credibility measure from fuzzy theory and a Non-dominated Sorting Genetic Algorithm II to solve the FCCP and stochastic multi-objective programming, respectively.

3.1. Transformation of fuzzy credibility constraints

Assuming that the triangular fuzzy variable expression is $\tilde{\xi} = (r^1, r^2, r^3)$, then for any real number r , the credibility of $\tilde{\xi} \leq r$ is:

$$Cr\{x \leq \tilde{\xi}\} = \begin{cases} 1, & x \leq b_1 \\ \frac{2b_2 - b_1 - x}{2(b_2 - b_1)}, & b_1 \leq x \leq b_2 \\ \frac{x - b_3}{2(b_2 - b_1)}, & b_2 \leq x \leq b_3 \\ 0, & x \geq b_3 \end{cases} \quad (17)$$

Generally, a meaningful credibility level λ should be greater than 0.5 (Khanjani Shiraz et al., 2017; Omrani et al., 2021). Based on the above formula, the fuzzy credibility constraints in the model, equations (8) and (10), can be reformulated as:

$$SW_{tn} + IW_{tn} + BW_{tn} \leq W_{m2} + (1 - 2\lambda)(W_{m2} - W_{m1}), 0.5 \leq \lambda \leq 1 \quad (18)$$

$$NW_{m1} + (1 - 2\lambda)(NW_{m2} - NW_{m1}) \leq JY_{tn} + IW_{tn}, 0.5 \leq \lambda \leq 1 \quad (19)$$

Where W_{t1} and W_{t2} represent the lower and upper limits of water resource supply. NW_{m1} and NW_{m2} represent the lower and upper limits

of crop water requirements, respectively.

3.2. Non-dominated Sorting Genetic Algorithm II

To address issues such as the poor distribution (clustering) of solutions along the Pareto front and insufficient convergence in traditional heuristic algorithms for multi-objective optimization, this study utilizes the Non-dominated Sorting Genetic Algorithm II (NSGA-II). Through non-dominated sorting, crowding distance comparison, and elitist preservation strategy, NSGA-II can efficiently generate Pareto-optimal solution sets with high diversity and precision, also making it suitable for complex scenarios of dynamic multi-objective nonlinear programming (López et al., 2021). The specific implementation steps are as follows (Fig. 2):

- (1) Encode decision variables (water volume allocated from reservoirs SW_{thin} , internal cultivated area M_{hm} , and external water diversion amount for irrigation areas BW_{thin} as continuous variables, and encode external water diversion decisions (0–1 variables) as binary variables, forming mixed-integer chromosomes (Zhou et al., 2021).
- (2) Use the four objective functions of the original model (maximizing economic benefits, maximizing clean energy production, minimizing water footprint, and minimizing carbon footprint) as independent optimization objectives for NSGA-II, avoiding bias caused by subjective weight settings, while adding constraints and FCCP.
- (3) Design genetic operations through Simulated Binary Crossover (SBX), bit-flip mutation, and elitist preservation strategy (Sutar and Jadhav, 2023; Brahami et al., 2022), while performing non-dominated sorting and selection to generate new populations. Continue population iterations until the maximum number of generations is reached.
- (4) To address the dynamic changes of agricultural seasons, divide the optimization process into multiple temporal stages, with each stage calling NSGA-II to generate the current resource allocation scheme, achieving global temporal optimization.

4. Case study

4.1. Study area

The Minjiang River Basin (MRB), as an important first-level tributary of the upper Yangtze River (Fig. 3), is selected as the case study area because it represents a typical WEFEN coordination challenge with clear hydrological boundaries. The basin is the core carrier of water supply for the Chengdu Plain and western Sichuan province (Li et al., 2024), spanning 750 km with an elevation difference of 3560 m and average annual runoff of 2910 m³/s, directly supporting industrial and agricultural production, energy development, and ecosystem stability throughout the region. The spatial distribution creates inherent challenges where upstream regions such as Aba Prefecture account for 65 % of average annual runoff while downstream Chengdu Plain's water

demand exceeds 70 % of the total, which can be exacerbated by seasonal variations.

To ensure result validity and avoid external regional influences, this study treats the MRB as a hydrologically self-contained system where water resources are predominantly sourced from internal precipitation and snowmelt. External influences are controlled through fixed boundary conditions including energy price parameters and predetermined inter-basin water transfer agreements, ensuring that optimization results reflect internal WEFEN coordination mechanisms rather than external dependencies. This boundary definition allows the model to focus on the most critical resource allocation decisions that basin managers can directly influence within the watershed.

The MRB faces multiple quantifiable challenges in water-energy-food systems security (Ju et al., 2023) that demonstrate the urgent need for integrated optimization. Extreme event vulnerabilities are exemplified by the 2022 high temperatures that caused river levels 40 % below average, triggering drought across 120,000 ha of farmland and 8 billion CNY in industrial losses (Emergency Management Department of Sichuan Provincial Government, 2022). The energy-water interdependency is evidenced by the basin's 4800 MW hydropower capacity (35 % of provincial total) experiencing 26 % efficiency decline during drought periods. Agricultural inefficiencies manifest through water consumption of 1200 m³/ton grain production (15 % above national average) and non-point source pollution contributing 42 % of the basin's carbon emissions ("Minjiang Ecological Blue Book," 2021). Ecosystem degradation includes 30 % wetland area reduction since the 1990s with ecological flow satisfaction rates below 60 %, obstructing fish migration channels (Minjiang Water Resources Commission, 2023). These challenges create complex trade-offs between hydropower generation, agricultural irrigation, carbon emission control, and ecological flow requirements that necessitate the proposed WEFEN optimization framework.

Based on our WEFEN framework, this study constructs a dynamic optimization model to provide a scientific paradigm for the MRB to achieve coordinated development of "efficient water resource utilization-secure energy supply-stable food production with increased income-carbon neutrality goals" through multi-objective collaborative optimization, helping to implement the ecological priority strategy of the Yangtze River Economic Belt.

4.2. Data description

Table 3 provides agricultural data regarding available land, grain yield per unit area, crop planting ratios, and water consumption per unit irrigated area. Table 4 includes carbon emission data and usage amounts of main raw materials, fertilizers, pesticides, and diesel in the agricultural production process (Qin et al., 2024; Liu et al., 2024). Table 5 shows the minimum water requirements for sectors in some regions and water requirements affected by uncertainty. All data sources used for the tables and for the case study are written below the tables, with data coming from official statistical yearbooks, official government reports, and expert websites. The total water resources and uncertainty disturbances for each region are calculated using the multi-year average

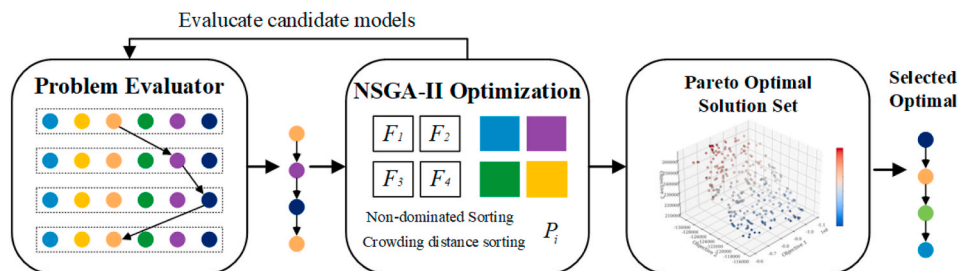


Fig. 2. NSGA-II algorithm flow chart.

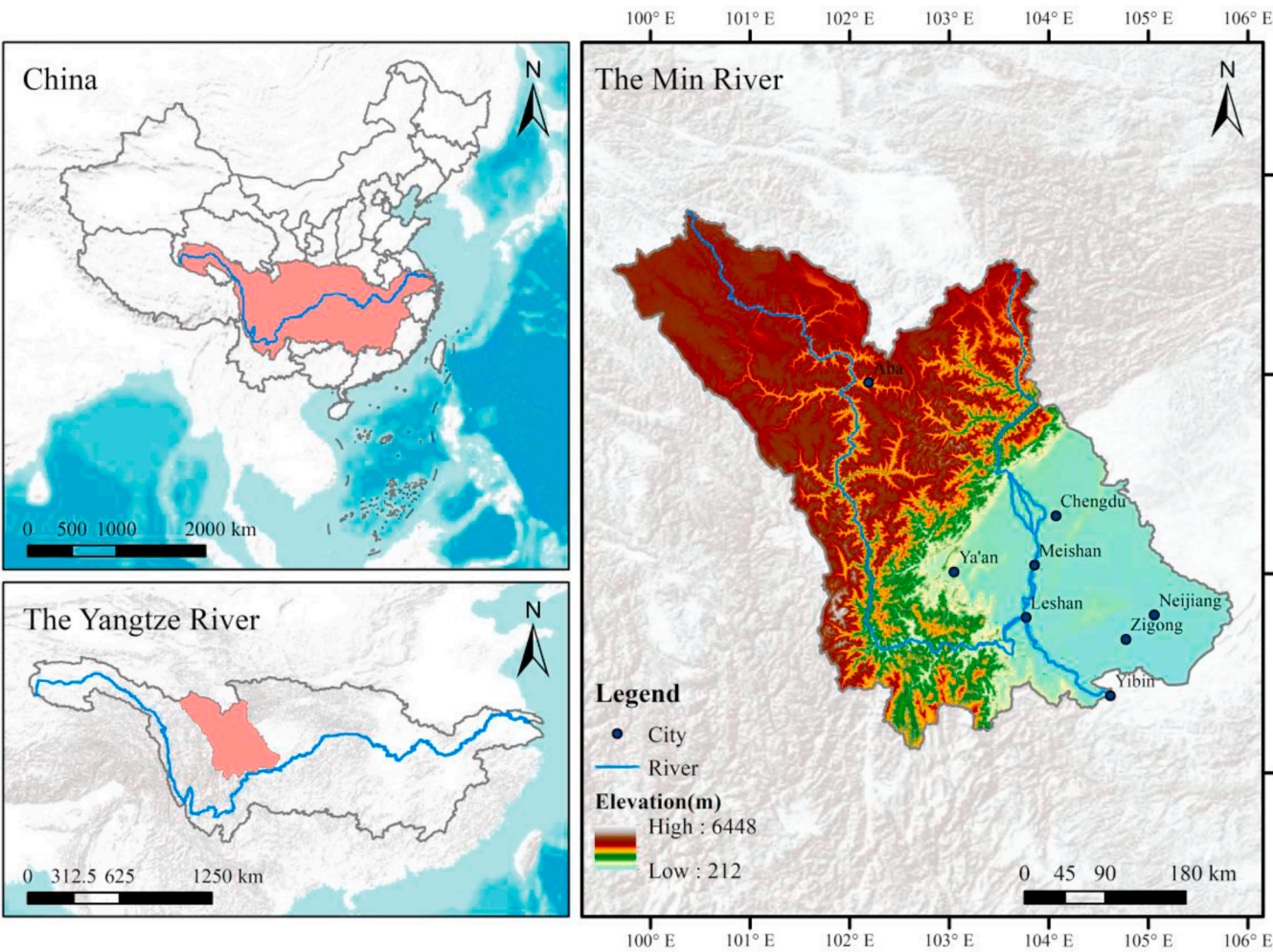


Fig. 3. Location map of the Minjiang river basin.

Table 3
Agricultural planting area, irrigation area, and yield data.

City	Minimum planting area (ha)	Maximum planting area (ha)	Ratio of irrigation area to planting area	Water consumption per irrigated ha (m ³)	Grain yield per unit area (kg/ha)
Chengdu	318.27	425.493	0.508	502.2380	5061.8
Zigong	166.93	213.083	0.272	260.7000	5037.0
Neijiang	221.39	266.498	0.278	228.3333	4659.7
Leshan	168.75	180.613	0.405	395.9009	4091.7
Meishan	226.40	256.364	0.534	380.9333	4858.3
Yibin	316.30	337.303	0.314	262.2064	4716.3
Ya'an	63.91	91.404	0.459	451.5656	4374.0
Aba	27.08	41.455	0.393	31.0083	3593.1

Data source: Sichuan Statistical Yearbook (2011–2020), Water Resources Bulletins of various cities (2011–2020)

Table 4
Carbon emission data for agricultural production.

Raw Material	Carbon Emission Factor CO ₂ e/kg	Usage
Fertilizer	0.8956 kg	20.75 kg/ha
Pesticide	4.9341 kg	1–1.5 L/ha
Diesel	2.68 kg	1.5–5 L/ha

Data source: Carbon Emissions, Carbon Absorption, and Carbon Footprint of Chinese Farmland Ecosystems <https://max.book118.com/>

values from 2011 to 2022.

4.3. Analysis of case study results

4.3.1. Analysis of NSGA-II optimization results

The solution is obtained using Python 3.9, with the population size set to 300 (popsize = 300) and the maximum number of iterations limited to 800 (n_gen = 800). A random initialization strategy is employed to enhance both the diversity of solutions and the stability of convergence (Zhang et al., 2023). As the evolutionary generations increase, the algorithm gradually converges to the Pareto frontier, which demonstrates the strong convergence capabilities of NSGA-II in addressing dynamic multi-objective optimization problems.

Fig. 4 illustrates the convergence curves of the objective function iterations for the four seasons: spring, summer, autumn, and winter. Overall, the convergence is faster in autumn and summer, primarily due to the abundance of resources, fewer optimization constraints, meaning less conflict between objectives. In contrast, convergence is slower in spring and winter, mainly because of more significant resource limitations, greater conflicts between optimization objectives, and higher uncertainty in the variables. Therefore, the availability of resources across different seasons plays a critical role in influencing optimization efficiency.

In terms of economic benefits (F1), the irrigation district achieved seasonal outputs of 1.42×10^8 CNY in spring, 2.66×10^8 CNY in summer, 12.43×10^8 CNY in autumn, and 1.07×10^8 CNY in winter. The markedly higher economic performance in autumn is primarily

Table 5Basic water demand data for various regions in the Minjiang river basin (m³).

City	Season	Minimum Domestic Water Demand	Minimum Agricultural Water Demand	Minimum Industrial Water Demand	Total Water Resources
Chengdu	Spring	6171.74	14289.77	3901.83	29212.90
	Summer	25699.50	59503.49	16247.47	121644.31
	Autumn	90032.64	208457.60	56919.51	426154.51
	Winter	11596.11	26849.14	7331.17	54888.29
Zigong	Spring	778.94	1973.69	731.58	4252.65
	Summer	4065.22	10300.38	3818.01	22193.88
	Autumn	8246.87	20895.76	7745.37	45023.41
	Winter	1708.97	4330.16	1605.04	9330.06
Nejiang	Spring	940.47	2185.52	785.52	4050.45
	Summer	7512.00	17456.88	6274.23	32352.82
	Autumn	6979.92	16220.38	5829.82	30061.24
	Winter	2167.60	5037.22	1810.45	9335.48
Leshan	Spring	685.65	2011.82	733.76	4378.49
	Summer	5005.52	14687.23	5356.78	31965.04
	Autumn	15022.69	44079.75	16076.92	95934.41
	Winter	2086.14	6121.19	2232.54	13322.04
Meishan	Spring	688.05	3144.84	726.93	5177.92
	Summer	3907.07	17857.71	4127.81	29402.31
	Autumn	11330.18	51785.97	11970.30	85264.43
	Winter	1774.70	8111.47	1874.96	13355.35
Yibin	Spring	1943.08	5108.43	2726.59	14771.65
	Summer	6367.27	16739.76	8934.73	48404.96
	Autumn	7316.61	19235.59	10266.85	55621.93
	Winter	2973.04	7816.22	4171.85	22601.48
Ya'an	Spring	392.78	1174.59	490.04	2480.13
	Summer	2494.08	7458.47	3111.65	15748.28
	Autumn	6299.03	18837.1	7858.79	39773.87
	Winter	1314.12	3929.84	1639.52	8297.70
Aba	Spring	250.76	467.31	80.83	1160.85
	Summer	2665.13	5024.41	873.81	12451.80
	Autumn	2415.18	4553.20	791.86	11284.03
	Winter	791.39	1491.97	259.48	3697.50

Data source: "Water Allocation Scheme for Major River Basins in Sichuan Province (2019)," "Sichuan Water Resources Bulletin 2011–2020".

attributed to abundant precipitation and high crop yields during the harvest period. For clean energy production (F2), the optimization algorithm converged most rapidly in autumn, as post-harvest straw resources were plentiful, significantly enhancing bioenergy conversion efficiency. In contrast, winter exhibited the smallest improvement in energy output, with only a 2.7 % increase, mainly due to limited straw residues and reduced hydropower availability resulting from low precipitation levels.

Regarding water footprint minimization (F3), the optimized values for spring, summer, and winter were reduced to $2.28 \times 10^5 \text{ m}^3$, $0.95 \times 10^5 \text{ m}^3$, and $1.85 \times 10^5 \text{ m}^3$, with reduction of 6.50 %, 9.49 %, 9.75 %, respectively. In autumn, although reservoir storage was sufficient during summer and considerable external water diversion was implemented, the increased water demand associated with high crop yields limited further reduction. As a result, the autumn water footprint decreased to $1.45 \times 10^5 \text{ m}^3$, representing a reduction of 17.24 %.

In terms of carbon footprint minimization (F4), optimization across all seasons led to significant reductions in agricultural carbon emissions, with final values reaching $1.34 \times 10^5 \text{ kg CO}_2\text{e}$ in spring, $1.33 \times 10^5 \text{ kg CO}_2\text{e}$ in summer, $1.37 \times 10^5 \text{ kg CO}_2\text{e}$ in autumn, and $1.32 \times 10^5 \text{ kg CO}_2\text{e}$ in winter, respectively with a reduction of 8.84 %, 8.90 %, 5.51 %, 9.20 %. Notably, autumn showed the smallest reductions, with the active agricultural outputs lead to smaller reduction in carbon emissions.

The distribution of the Pareto front illustrates the relationships between different objectives in Fig. 5. Economic benefits (F1) and clean energy (F2) exhibit partial negative correlation in summer and autumn, indicating conflict between enhancing agricultural economic benefits and increasing clean energy production through water resource allocation. However, this correlation is weaker in spring and winter, as water resources in spring are primarily directed towards agricultural irrigation

demands, and winter scarcity of hydropower resources limits clean energy production. Across all seasons, water footprint (F3) and carbon footprint (F4) show strong positive correlation, primarily because both are influenced by planting area size. Larger planting areas lead to higher agricultural irrigation and fertilization demands, thus increasing both water and carbon footprints. Clean energy (F2) and water footprint (F3) exhibit non-linear negative correlation for all seasons, as maximizing clean energy output leads to higher water consumption. The relationship between economic benefits (F1) and carbon footprint (F4) is less clear, particularly in spring and winter, suggesting that the balance between economic growth and carbon emissions is influenced by various factors, such as water resource scheduling and clean energy proportion. In autumn and summer, partial correlation is observed, indicating that when water resources are sufficient, optimizing water resource allocation can reduce carbon emissions while supporting economic growth.

Fig. 6 uses a three-dimensional visualization of the Pareto frontier to illustrate the nonlinear characteristics of the optimization space. With F1, F2, and F3 as the axes and F4 represented by a color gradient, points on the Pareto frontier indicate that the optimal solution set forms various trade-offs between the different objectives. The optimal solution set reveals various trade-off scenarios between different objectives. The color gradient transitions from blue (indicating low carbon emissions) to red (indicating high carbon emissions). One clear observation is that some solutions represent enhance both economic benefits and clean energy output while simultaneously causing a significant increase in carbon emissions. This particularly highlights the need for policymakers to prioritize low-carbon development strategies in the optimization process as without intervention, higher economic benefits can be achieved but at the cost of the environment (water and carbon footprints).

Clearly, as F2 (clean energy output) (F2) increases, the points on the Pareto front become bluer, which is linked to clean energy production, and is true across all seasons. An interesting and related point to raise is that it is possible to achieve high economic benefits (F1) even under lower carbon emissions. Specifically, high economic benefits can be obtained in spring, summer, and winter without the need for high carbon emissions. Given that the main harvesting season is in autumn, it is natural that only in autumn there is a negative trade-off between clean energy and economic benefits. In other words, the returns to production are so much higher in autumn that trying to promote clean energy can only hamper economic benefits.

4.3.2. WEFEN system optimization results

We adopt a normalization technique plus a weighted method for decision-making to screen for the optimal solution (Emamjomehzadeh et al., 2024). The objective values are processed using min-max normalization, mapping all objective values to the [0,1] range, with the calculation formula as follows:

$$F_i^{\text{norm}} = \frac{F_i - F_i^{\min}}{F_i^{\max} - F_i^{\min} + \epsilon} \quad (17)$$

where $F_i^{\min}$ and $F_i^{\max}$ are the minimum and maximum values of objective i , and ϵ is a very small number (such as $1\text{e-}9$) used to prevent division by zero. After normalization, a weight vector $\mathbf{w} = (w_1, w_2, \dots, w_m)^T$ is set according to the importance of different objectives, and the weighted objective value for each Pareto solution is calculated:

$$S = \sum_{i=1}^m w_i F_i^{\text{norm}} \quad (18)$$

The solution with the minimum weighted sum is selected as the optimal solution, with the values of the decision variables constituting said solution. In the subsequent analysis, we use this solution which allows us to propose an overall plan for water resource allocation, energy, food, and ecological sustainability in the MRB.

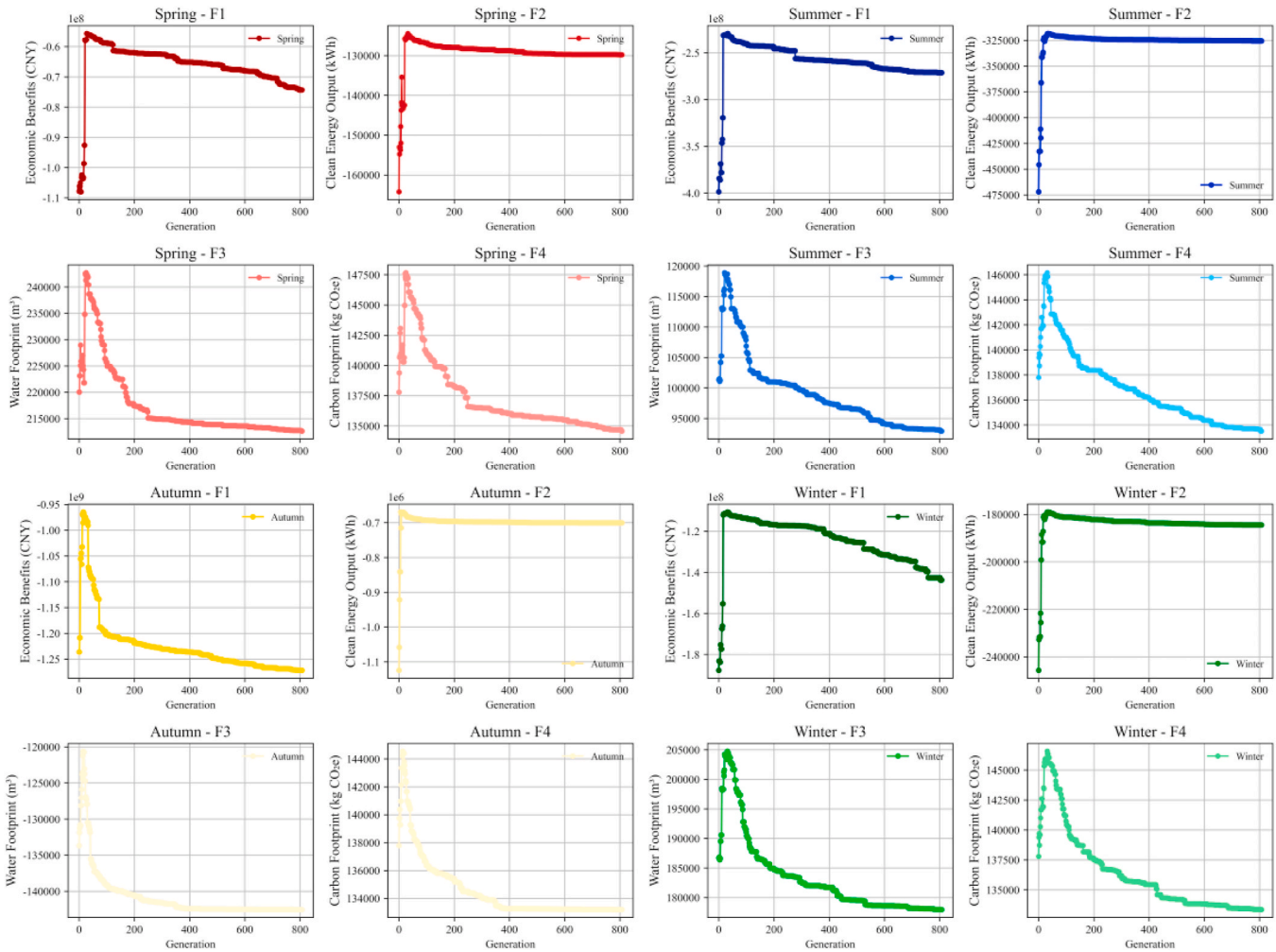


Fig. 4. Four seasons iteration convergence graphs.

(1) Water Resource Allocation

Figs. 7 and 8 present the seasonal patterns of water storage, external water diversion, and supply-demand gaps across various cities. The findings reveal that water demand during summer and autumn is significantly higher than in spring and winter. Despite higher precipitation levels in summer and autumn, natural supply still falls short of meeting demand, leading to a marked increase in external water diversion during autumn. In particular, Leshan and Meishan reported the highest diversion volumes, reaching $3.25 \times 10^5 \text{ m}^3$ and $3.30 \times 10^5 \text{ m}^3$, respectively—substantially exceeding those of other cities. In winter, reduced precipitation results in generally lower reservoir storage levels. Therefore, it is recommended to minimize reservoir releases during this season and prioritize water retention to ensure a stable water supply for agricultural and urban needs in the following year.

Notable spatial disparities exist in urban water supply and demand which is true across all seasons. Cities such as Chengdu, Yibin, Meishan, and Leshan have substantially higher water demands than areas like Aba. This is primarily due to their more developed economies and strong industrial water consumption, especially for hydropower generation, which increases the likelihood of water shortages. It is advisable that these cities moderately increase reservoir releases during summer and autumn to meet essential agricultural, industrial, and domestic water needs in the other seasons.

(2) Clean energy output

Urban clean energy mainly includes hydropower and biomass energy output. Fig. 9 shows that, except for Chengdu, biomass energy output in spring is generally higher than hydropower output. Yibin City has the highest hydropower output at $0.82 \times 10^5 \text{ kWh}$. Yibin, which is located at the confluence of the Jinsha River and Minjiang River basins, possesses abundant hydropower resources. Hydropower output reaches its annual peak in summer, with Chengdu's hydropower generation reaching $3.38 \times 10^5 \text{ kWh}$, far exceeding other cities. During the summer season, abundant precipitation enables hydropower generating units to operate at full capacity, resulting in a total hydropower output of $1.44 \times 10^5 \text{ kWh}$ across the basin. Additionally, large-scale hydropower stations in cities such as Yibin and Leshan exhibit strong generation capacities. Given the high electricity demand in summer, their dispatching flexibility and operational efficiency are significantly improved.

Biomass energy output reaches its peak in autumn, with Chengdu, Yibin, and Meishan reaching $0.48 \times 10^5 \text{ kWh}$, $0.14 \times 10^5 \text{ kWh}$, and $0.10 \times 10^5 \text{ kWh}$, respectively. Autumn is the agricultural harvest season with abundant straw resources, significantly increasing biomass energy production. Additionally, increased demand for agricultural waste processing results in higher use of biomass energy. Hydropower output decreases substantially in winter, with Yibin's hydropower generation dropping to $0.14 \times 10^5 \text{ kWh}$. Reduced precipitation in winter results in lower reservoir levels, decreasing available water for hydropower generation. Additionally, hydropower stations intentionally conserve water during winter to ensure sufficient supply for spring and summer, further reducing winter hydropower output.

Biomass energy accounts for significantly higher proportions in cities

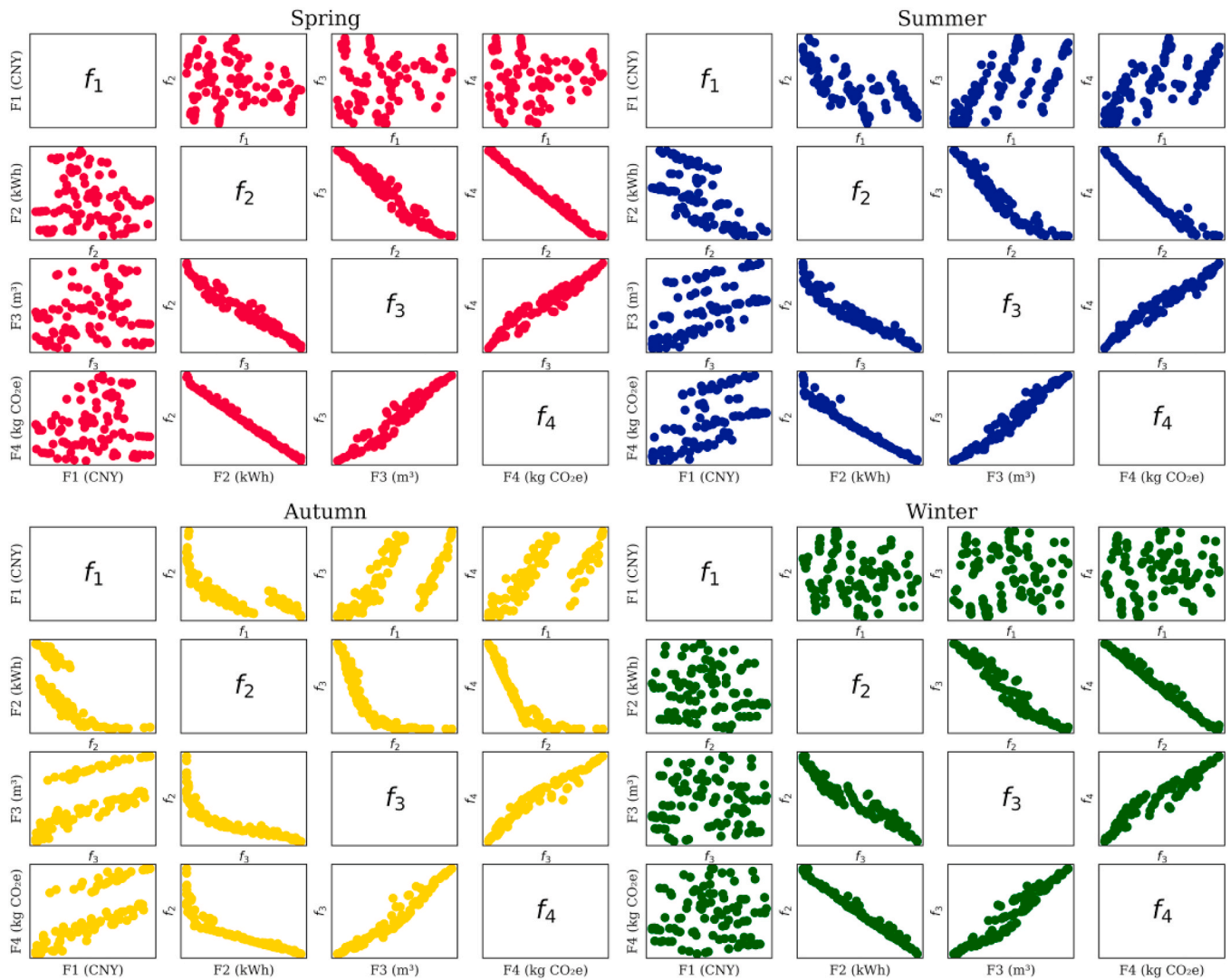


Fig. 5. Four seasons objective function scatter matrix diagrams.

such as Chengdu, Yibin, and Meishan, reaching 0.15×10^5 , 0.14×10^5 , and 0.10×10^5 kWh, respectively. These regions have large agricultural planting areas with abundant crop straw resources, but local water resources are relatively limited, and hydropower generation capacity is weaker, resulting in higher proportions of biomass energy use. As such, technologies that enhance the efficiency of converting agricultural waste would be particularly valuable.

The clean energy output efficiency per unit of water resources, as shown in Fig. 10, exhibits significant seasonal differences. Spring energy output efficiency is relatively high, with Zigong, Neijiang, and Meishan reaching 2.96, 3.32, and 2.99 kWh/m³, respectively. Clean energy output efficiency in winter is relatively low, with Chengdu at only 0.15 kWh/m³. Chengdu's annual demand reaches 53.58×10^8 m³, experiencing year-round water tension, with relatively less water resources available for hydropower generation. Autumn clean energy efficiency generally consistent across regions, ranging between 0.5 and 0.95 kWh/m³. After the autumn crop harvest, increased straw resources enhance biomass energy. Summer energy efficiency is relatively high but shows significant regional differences, with Yibin, Leshan, Meishan, and Neijiang reaching 0.6 kWh/m³, while Aba is only at 0.18 kWh/m³. This mainly reflects differences in regional economic development and water resource utilization methods. Upstream areas such as Aba and Ya'an have abundant water resources, with annual average runoffs of 2.85×10^8 m³ and 6.6×10^8 m³, but since water resources are primarily used

for agricultural irrigation (accounting for 58.29 % and 54.25 %, respectively), the clean energy conversion efficiency per unit of water resources is relatively low. In contrast, Yibin and Leshan have ports with more mature hydropower generation technology, with a total installed capacity of 6.4×10^6 kWh. Therefore, the clean energy conversion efficiency per unit of water resources is significantly higher.

(3) Food Output and Agricultural Economic Benefits

The results of Table 6 and Fig. 11 indicate that downstream cities such as Chengdu, Meishan, and Yibin have higher food output due to large cultivated land areas and well-developed agricultural infrastructure. Their annual food outputs are 64.45×10^6 tons, 44.17×10^6 tons, and 59.67×10^6 tons, with cultivated land areas of 1.28×10^3 ha, 0.93×10^3 ha, and 1.31×10^3 ha, respectively. Upstream areas such as Aba and Ya'an, constrained by higher-altitude climate, geographical conditions, and limited arable land resources, have significantly lower food output. Aba's total output across the four seasons is only 4.57×10^6 tons, and Ya'an is 11.23×10^6 tons, mainly limited by smaller cultivated land areas (0.12×10^3 ha and 0.26×10^3 ha, respectively) and poor agricultural production conditions. Upstream areas have less arable land due to cold climate and poor soil, with shorter crop growing cycles which limits food production.

From an economic benefit perspective, agricultural income is

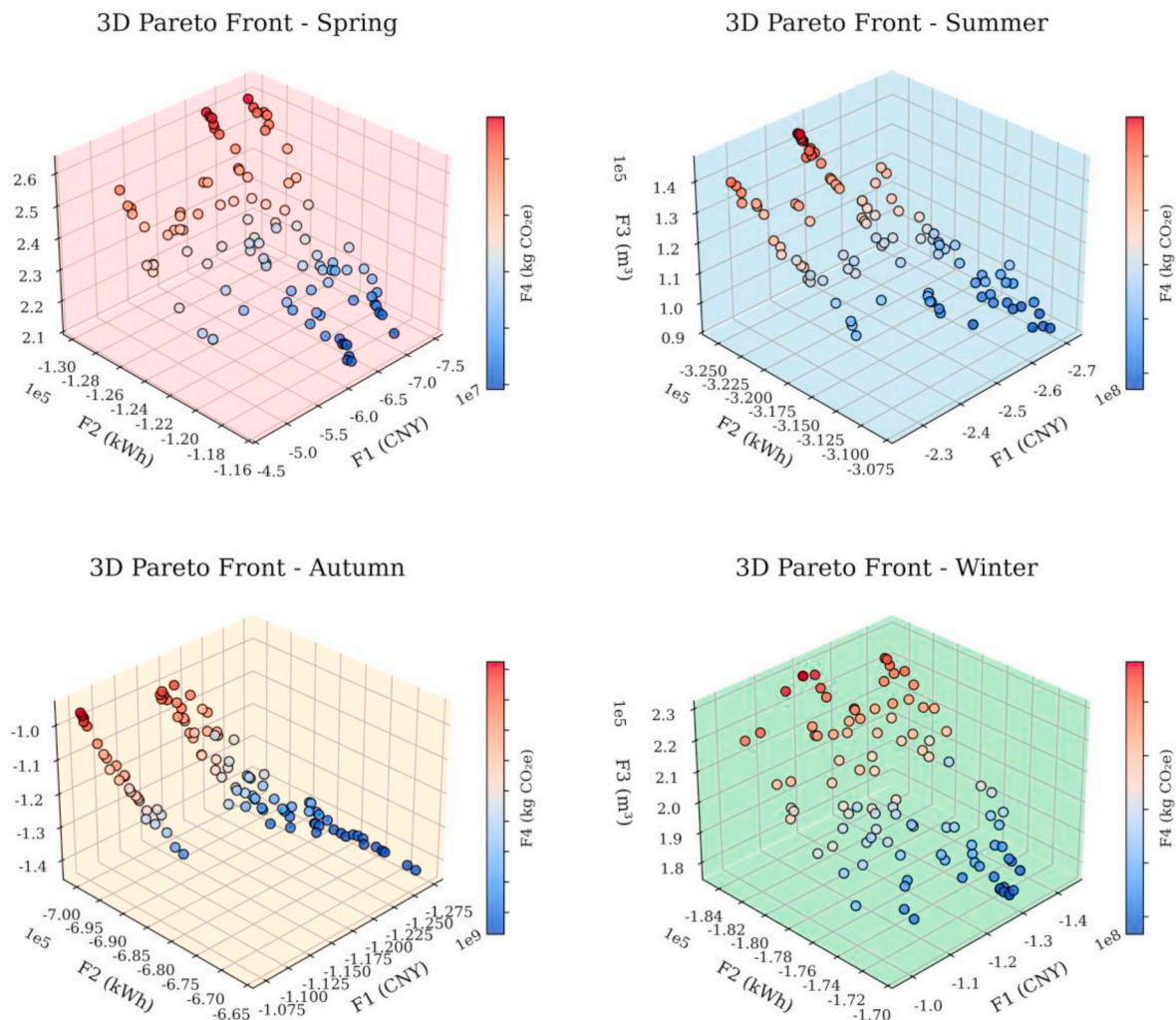


Fig. 6. Three-dimensional pareto frontier diagrams.

influenced by both food output and market prices. Cities such as Chengdu, Yibin, and Meishan make relatively larger agricultural economic contributions due to their larger planting scales and stable food market supply. For plateau regions like Aha, due to low food production, local governments may need to rely on specialty agriculture, agricultural subsidies, or external food imports to meet local food demand.

(4) Carbon Footprint, Water Footprint, and Environmental Impact

The main contributors of the carbon footprint in each city are fertilizers, pesticides, and diesel, with fertilizer carbon emissions accounting for the highest proportion. As seen in Fig. 12, Chengdu and Yibin have the highest total carbon emissions: Chengdu's annual agricultural carbon footprint is approximately 2.91×10^4 kgCO₂e, with fertilizer, pesticide, and diesel carbon footprints of 2.37×10^4 kgCO₂e, 0.14×10^4 kgCO₂e, and 0.41×10^4 kgCO₂e, respectively; Yibin's annual agricultural carbon footprint is approximately 2.73×10^4 kgCO₂e, with fertilizer, pesticide, and diesel carbon footprints of 2.35×10^4 kgCO₂e, 0.12×10^4 kgCO₂e, and 0.25×10^4 kgCO₂e, respectively. Aha has the lowest carbon emissions at only 0.27×10^4 kgCO₂e, which aligns with expectations, as it is a region with a relatively low industrial activity and a stronger emphasis on environmental sustainability. Chengdu and Yibin have larger cultivated land areas (1.27×10^3 ha and 1.26×10^3 ha, respectively) and rely heavily on agricultural machine, resulting in higher fuel consumption and increased carbon emissions. In contrast,

Aha's cultivated land area is only 0.13×10^3 ha, with agricultural production predominantly using small-scale traditional cultivation, with less dependence on machinery, fertilizers, and pesticides, all of which lead to a significantly smaller carbon footprint.

Carbon emissions data indicates that fertilizer usage accounts for approximately 70–86 % of emissions, while pesticide usage accounts for around 4–5 %. Nitrogen fertilizer production involves energy-intensive processes, resulting in significant carbon dioxide emissions. Additionally, the application of nitrogen fertilizers indirectly increases carbon emissions through ammonia volatilization and nitrification. Diesel-related carbon emissions depend on agricultural machinery use; diesel emissions in Chengdu, Meishan, and Yibin reached 4.11×10^3 kgCO₂e, 3.60×10^4 kgCO₂e, and 2.54×10^3 kgCO₂e, respectively, while regions like Aha emitted only 240.15 kgCO₂e due to lower usage of agricultural machinery.

Therefore, from a spatial distribution perspective, agricultural carbon footprints show higher levels downstream and lower levels upstream. Cities located in the middle and lower reaches, such as Chengdu, Yibin, and Leshan, have higher total carbon emissions, while upstream areas such as Aha and Ya'an have lower carbon footprints. Carbon emissions should be reduced by optimizing fertilizer use, decreasing excessive application of nitrogen fertilizers, optimizing agricultural production, and encouraging upstream regions (such as Aha and Ya'an) to adopt low-carbon agricultural methods. These recommended measures promote the development of agricultural production whilst being ecologically sustainable.

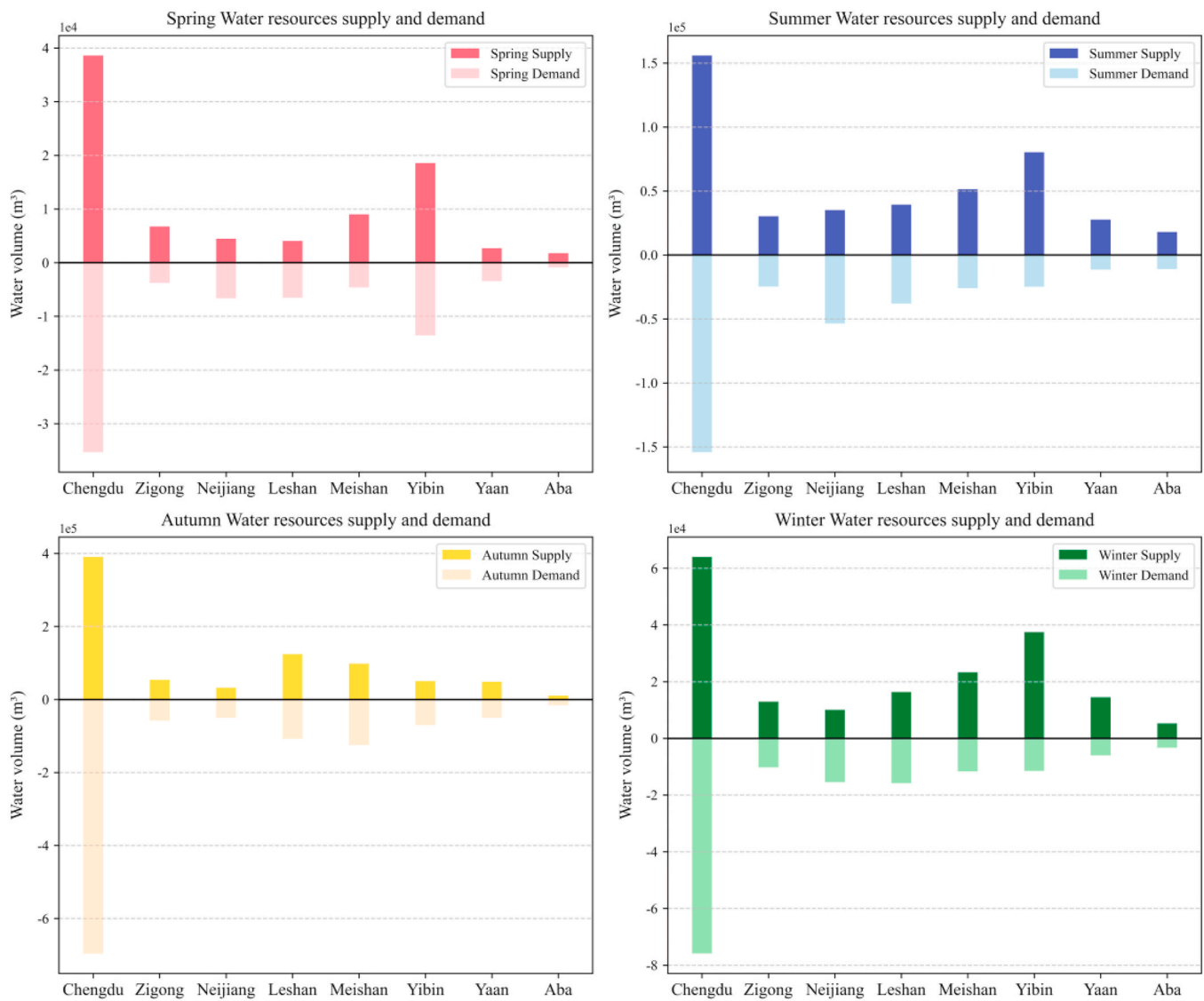


Fig. 7. Proportions of water resource supply and demand in 8 cities across four seasons.

As shown in Fig. 13, Chengdu's blue water footprint reaches its peak in autumn at $1.27 \times 10^5 \text{ m}^3$. Crops typically reach maturity in autumn, increasing water demand, with concurrent high industrial and domestic water demand, significantly increasing agricultural irrigation's dependency on surface water and groundwater. Aba's blue water footprint is lower with smaller annual fluctuations, with a winter blue water footprint of only $0.10 \times 10^5 \text{ m}^3$, due to the region's smaller agricultural production scale, abundant water resources, and limited development. The green water footprint shows the contribution of precipitation to agricultural production. Yibin, Leshan, and Meishan have the highest green water footprints, reaching $0.19 \times 10^6 \text{ m}^3$, $0.27 \times 10^6 \text{ m}^3$, and $0.46 \times 10^6 \text{ m}^3$ in autumn, respectively. These cities rely more on natural precipitation during seasons with sufficient rainfall, reducing demand for irrigation water and improving the efficiency of water resource use.

The grey water footprints reflect agricultural pollution associated with fertilizers and pesticides, and represents the volume of freshwater required to dilute these pollutants to acceptable levels. Chengdu's grey water footprint reaches its highest in autumn at $5.27 \times 10^5 \text{ m}^3$, decreasing to $5.07 \times 10^5 \text{ m}^3$ in winter, possibly due to high crop yields in autumn requiring larger inputs of fertilizers and pesticides, leading to higher water pollution. Aba's grey water footprint is lower, ranging between 442.15 and 648.16 m^3 throughout the year, meaning less

agricultural pollution, as indicated by smaller-scale agricultural activities, and more ecologically friendly production methods that cause less water pollution.

From both a temporal and spatial view of the water footprints, Chengdu and Yibin have significantly higher total water footprints than other cities, especially reaching peaks in autumn and summer at $5.27 \times 10^5 \text{ m}^3$ and $5.21 \times 10^5 \text{ m}^3$, respectively. Cities with higher green water footprints (such as Yibin, Leshan, and Meishan) can better utilize precipitation to reduce their reliance on blue water consumption, supporting the sustainable use of water resources across the watershed. Upstream areas such as Aba have less agricultural production and lower water resource consumption, meaning their lower green water footprint indicates potential for improving agricultural development through better crop selection, land use planning, and more effective use of natural precipitation.

4.3.3. Credibility level sensitivity analysis

Due to uncertainty in regional rainfall-runoff (i.e. different water inflow levels), this study integrates the probabilistic nature of water availability into our dynamic optimization model. Specifically, the most likely value of the triangular fuzzy number is set as the average annual available water volume, while the minimum and maximum values are

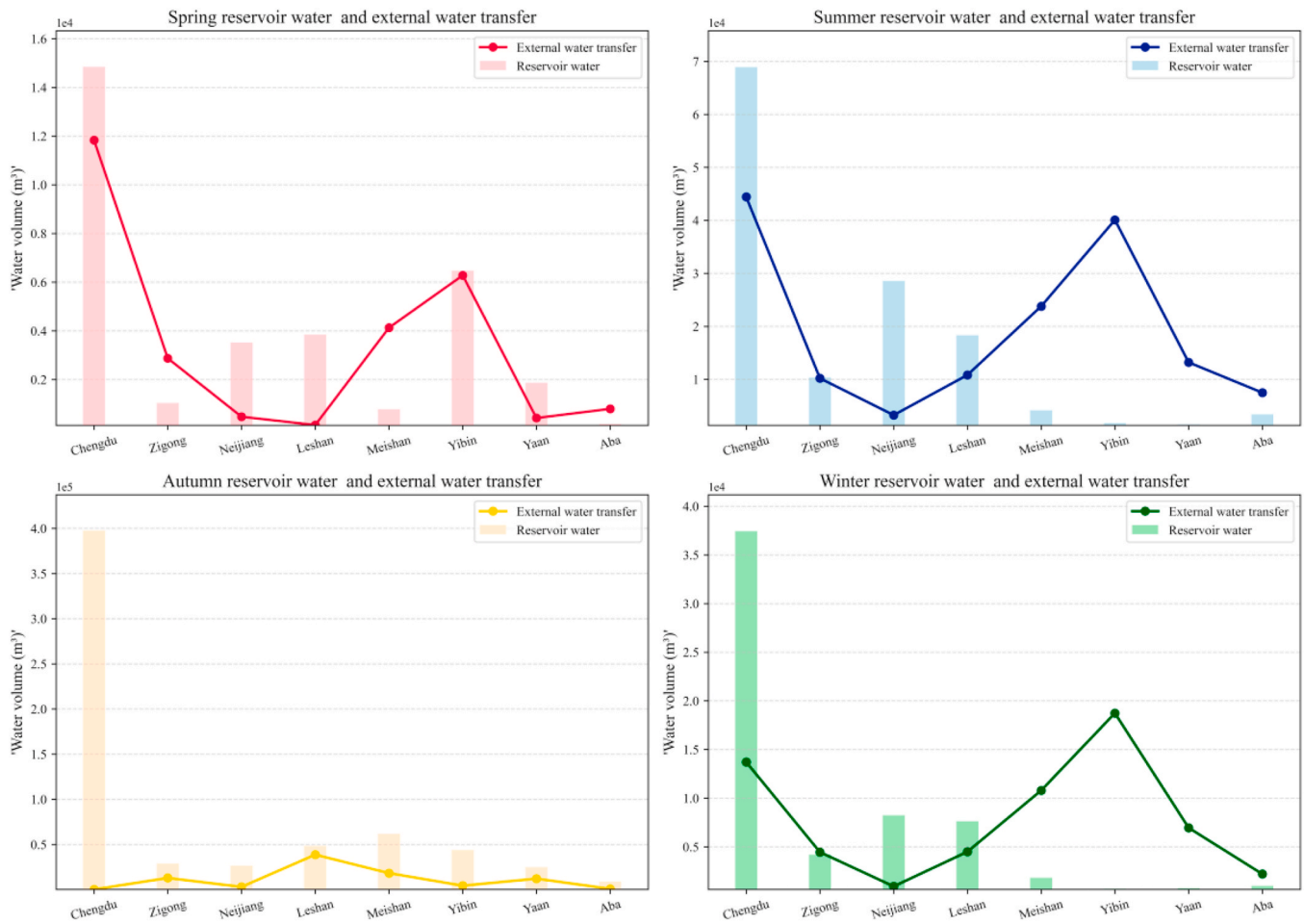


Fig. 8. Water storage and external water diversion volumes for 8 cities across four seasons.

defined using the 95 % confidence interval calculated based on historical data using interval estimation methods. According to existing research the credibility level in fuzzy analysis is typically required to exceed 0.5 (Peykani et al., 2022). Therefore, this study selects a credibility confidence level range from 0.5 (constraint not satisfied) to 0.95 (constraints almost completely satisfied), with a step size of 0.05 for sensitivity analysis, recalling that when λ is higher/lower there is lower/higher tolerance for uncertainty.

To adapt to the standard framework of PyMOO (Python package for NSGA-II) in which all objectives are minimized, F1 (economic benefits) and F2 (clean energy) are multiplied by -1 in the computations. Therefore, a lower value on the vertical axis in the figure corresponds to higher actual values (economic benefits or clean energy output). As F3 (water footprint) and F4 (carbon footprint) are already objectives to be minimized, these objectives are maintained in their stated form.

As shown in Fig. 14, the curves for each λ exhibit substantial fluctuations in the initial generations (around generations 0–50), followed by gradual stabilization and convergence after approximately 100 generations. The curves become relatively stable between generations 600–800, indicating that the algorithm has identified a relatively optimal feasible solution during this phase. There are noticeable differences in the final convergence values for different λ values, suggesting that decision-makers with varying confidence levels regarding supply-demand balance (λ values) will obtain different trade-off combinations of economic benefits, clean energy output, water footprint, and carbon footprint. Specifically:

- (1) Economic Benefits (F1): From the final convergence points, the curves corresponding to $\lambda = 0.50$, $\lambda = 0.65$, and $\lambda = 0.85$ achieve

higher economic benefits, reaching 1.07×10^8 , 1.08×10^8 , and 1.05×10^8 CNY, respectively. In contrast, the economic benefits are lowest for $\lambda = 0.60$, $\lambda = 0.75$, and $\lambda = 0.90$, with only 0.95×10^8 CNY. Overall, the final economic benefits for different λ values range from approximately 0.95×10^8 to 1.08×10^8 CNY with lower λ values tending to yield higher economic benefits.

- (2) Clean Energy Production (F2): After approximately 100 generations, the curves gradually stratify and converge, with clean energy output decreasing as λ increases. For $\lambda = 0.60$ and $\lambda = 0.50$, clean energy production is higher, reaching 1.29×10^5 kgCO₂e and 1.28×10^5 kgCO₂e, respectively. At $\lambda = 0.70$ and above, clean energy output decreases to around 1.27×10^5 kgCO₂e.
- (3) Water Footprint (F3) and Carbon Footprint (F4): In general, both decrease steadily as λ increases, indicating that more conservative water resource allocation schemes help reduce water consumption and carbon emissions in the production process. For instance, when $\lambda = 0.8$, both the water footprint and carbon footprint reach their minimum values, at 2.11×10^5 m³ and 1.34×10^5 m³, respectively. Compared to the scenario with $\lambda = 0.6$, where the values are 2.15×10^5 m³ and 1.35×10^5 m³, these values are reduced by approximately 2 %. This demonstrates that stricter constraints on water resource utilization can reduce water waste and carbon emissions, reflecting improved sustainability characteristics. It is important to note that under the strictest water resource constraint ($\lambda = 0.9$), the water and carbon footprints are not at their lowest. This is due to the introduction of a significant amount of external water diversion to meet the constraints, which, in turn, negatively impacts the environment.

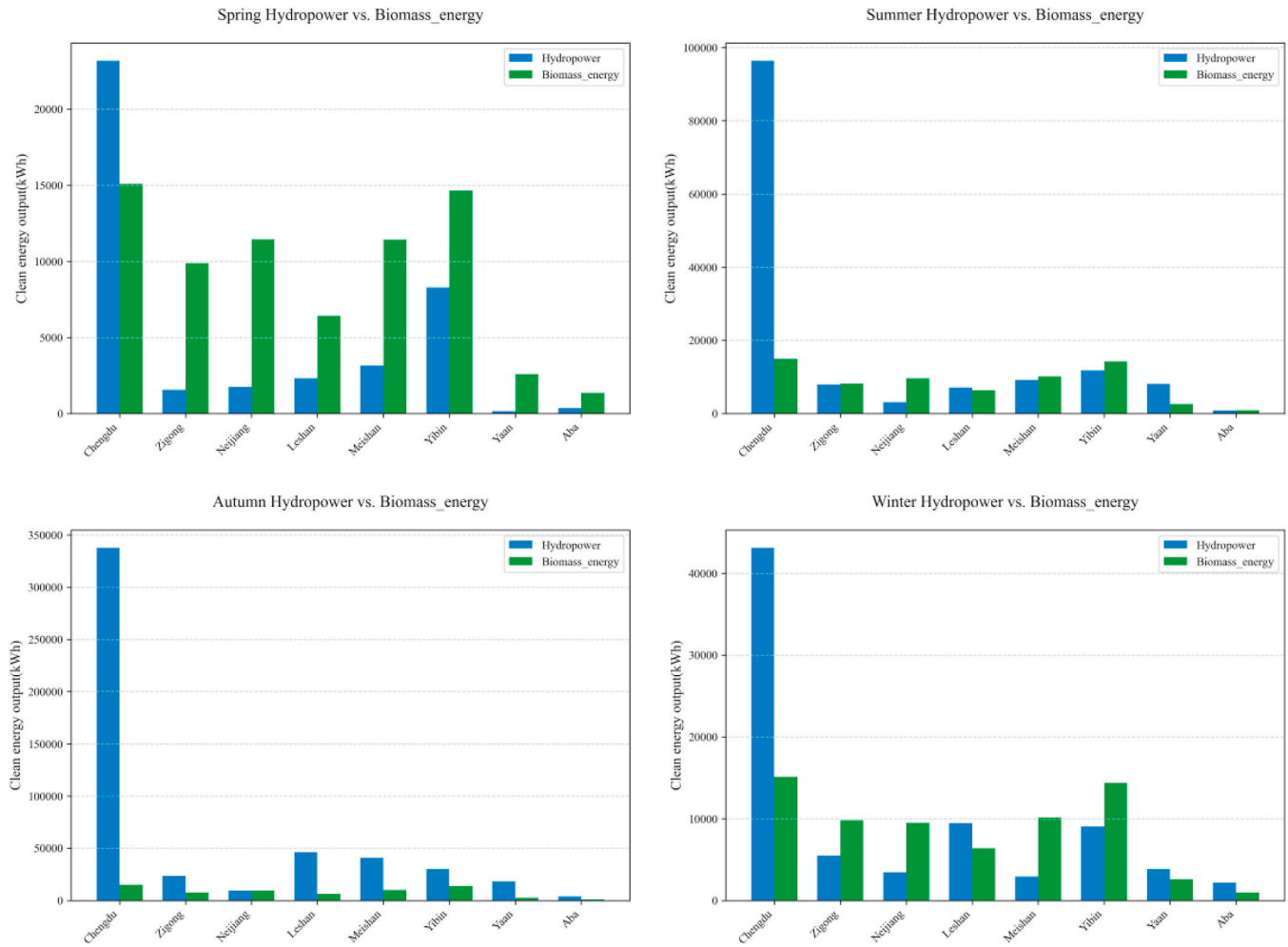


Fig. 9. Clean energy output across four seasons.



Fig. 10. Clean energy output efficiency across four seasons.

Thus, it can be observed that a higher level of confidence in resource allocation can, to some extent, sacrifice economic benefits to significantly reduce the risks associated with water scarcity and environmental pressure. Conversely, although lower levels of confidence may yield higher returns, they introduce considerable uncertainty regarding resource supply security.

Table 6

City food output and cultivated land area.

City	Total food output across four seasons (10^4 tons)	Total cultivated land area across four seasons (ha)
Chengdu	6448.82	1274.47
Zigong	3392.89	674.53
Neijiang	4128.18	887.78
Leshan	2784.68	680.85
Meishan	4415.88	910.49
Yibin	5966.63	1266.80
Ya'an	1122.39	256.84
Aba	449.43	125.19

Fig. 15 illustrates the changes in decision variables, including the water volume allocated to the reservoir (SW_{thn}), the internal cultivated area (M_{hm}), and the external water diversion volume for the irrigation district (BW_{thn}) under different confidence levels. As λ increases from 0.5 to 0.9, the reservoir allocation initially fluctuates significantly but gradually stabilizes, ultimately reaching around $0.63 \times 10^5 \text{ m}^3$. Under high confidence ($\lambda = 0.9$), the model tends to adopt a more “conservative” reservoir scheduling strategy.

The variation in cultivated area is relatively small, with the maximum fluctuation within the range of $\lambda = 0.5$ to $\lambda = 0.9$ being only 6.8 %. This indicates that the model consistently guarantees food security and basic irrigation needs, without making large-scale adjustments to the cropping structure, thereby ensuring the minimum requirements of the food security threshold. The external water

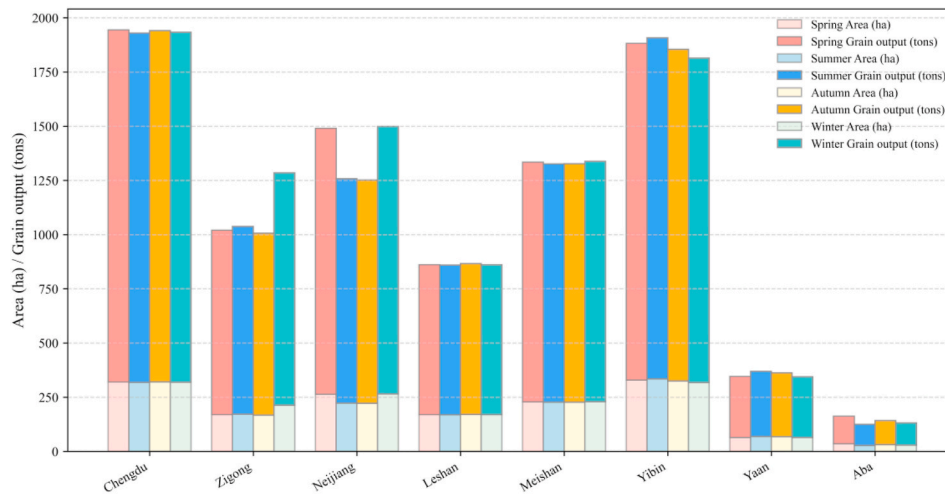


Fig. 11. Food output and cultivated land area for 8 cities across four seasons.

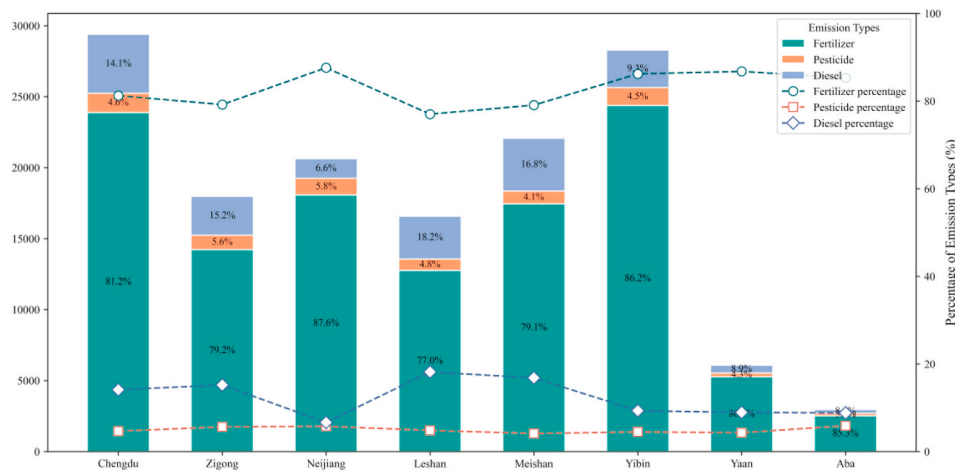


Fig. 12. Carbon footprint composition of cities in the MRB.

diversion volume (BW_{thn}) increases with fluctuations as λ rises. When $\lambda = 0.90$, the external water diversion volume reaches $1.44 \times 10^5 \text{ m}^3$, which is an increase of approximately 60.6 % compared to the $\lambda = 0.5$ scenario. This suggests that to meet the supply-demand balance required by the high-confidence target, the system requires more external water sources, resulting in higher water diversion costs.

Figs. 16 and 17 illustrate the distribution of the Pareto front for the objectives F1 versus F2 and F3 versus F4 under different λ values. From Fig. 16, it is evident that the Pareto solution sets for low λ (0.50–0.60) are somewhat more dispersed, indicating that with higher risk tolerance, the decision-maker generates more extreme solutions that prioritize either maximizing economic benefits or clean energy production. In contrast, under high λ (0.80–0.90), the solution sets (brown and light green) become more concentrated, with a narrower distribution range. This reflects the more stringent “stable demand satisfaction” constraint, suggesting that the model adopts a more conservative approach to avoid the risk of water resource shortages. From Fig. 17, the trade-off (in terms of solutions) between carbon and water footprints is clearly positively correlated with adjustments between F3 and F4 occurring within a relative tight interval (especially when compared with Fig. 16) as λ varies.

It is clear that different credibility levels λ have important implications on water resource allocation strategies, economic returns, and environmental sustainability, summarized as follows:

(1) When $\lambda = 0.70$ –0.80, a relative balance can be achieved between

economic benefits and environmental objectives such as water footprint and carbon footprint. (2) When $\lambda < 0.7$, the highest economic benefits may be obtained, potentially increasing by about 7 %, but correspondingly, the risks of water resource shortages or supply-demand mismatches are greater. (3) When $\lambda > 0.8$, shortage risks can be reduced and environmental indicators improved, but external water diversion volume and unit water supply costs increase by approximately 6.8 %, while economic benefits decrease by 4.7 %.

In terms of policy, if there is a focus on pursuing economic benefits with a reasonable risk tolerance for supply-demand mismatches, $\lambda \approx 0.70$ is a reasonable credibility level. If stability of water resource supply and environmental sustainability are of primary concern then credibility levels in the range of $\lambda = 0.80$ –0.90 are more appropriate. Overall, $\lambda = 0.80$ offers a balance between economic and environmental effects and is more appropriate for areas with higher requirements for water ecological protection, taking into account both supply-demand balance and sustainable development. Therefore, in the case analysis above, this paper uses $\lambda = 0.80$ as the baseline scenario for water resource allocation and effect evaluation.

4.3.4. System response to uncertainty analysis

Based on the credibility level sensitivity analysis results, different objective functions of the system exhibit distinct sensitivity characteristics to uncertainty. Economic benefits are most sensitive to uncertainty changes. When λ increases from 0.50 to 0.90, economic benefits

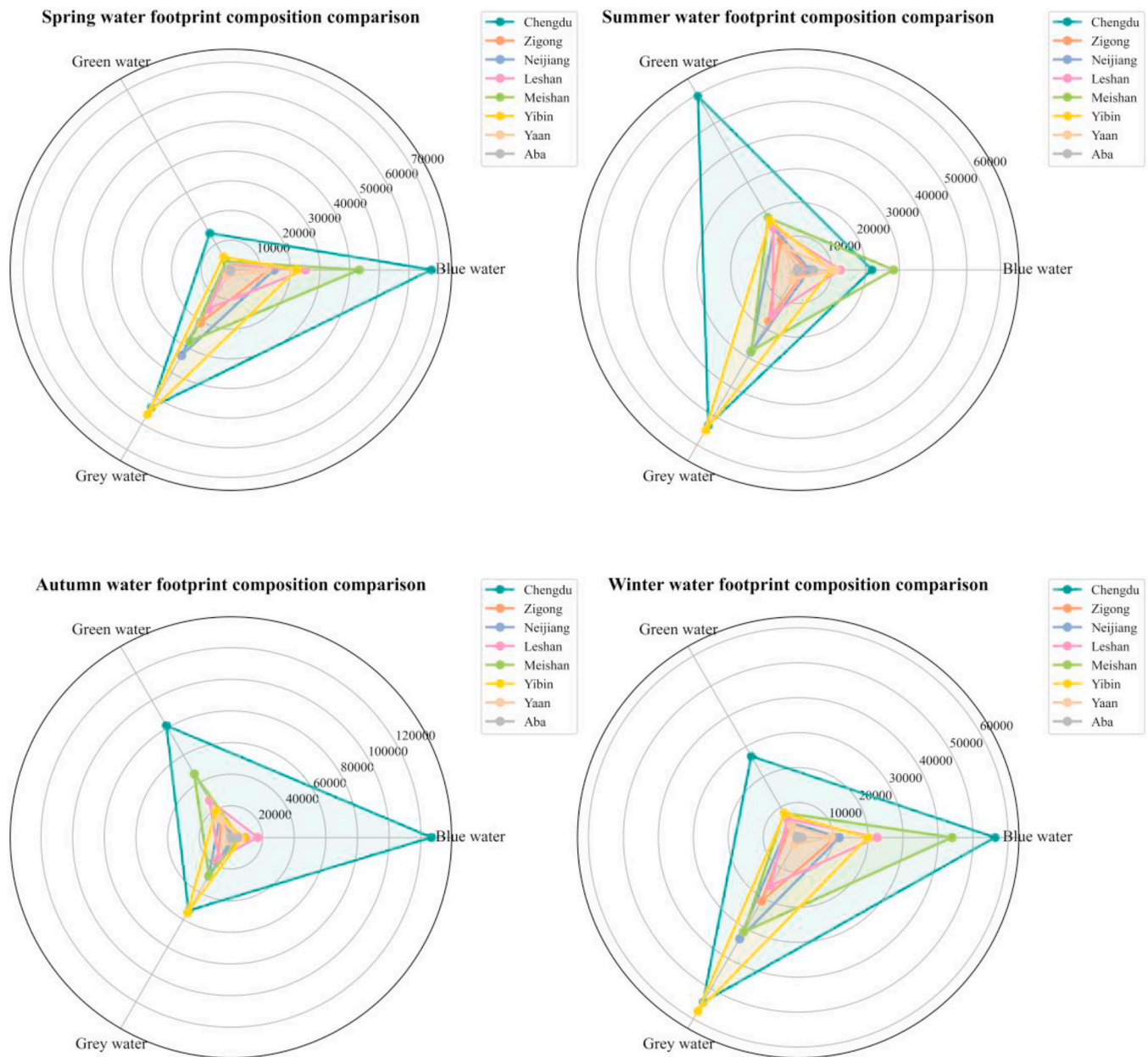


Fig. 13. Radar charts of water footprints for 8 cities across four seasons.

fluctuate from the highest value of 1.08×10^8 CNY to the lowest value of 0.95×10^8 CNY, representing a variation range of 12.0 %. Clean energy production maintains relatively high levels at $\lambda = 0.60$ and $\lambda = 0.50$ (1.29×10^5 and 1.28×10^5 kWh respectively), but decreases to approximately 1.27×10^5 kWh when λ exceeds 0.70.

In contrast, environmental objectives demonstrate stronger robustness. The water footprint varies by only 1.9 % from $\lambda = 0.6$ to $\lambda = 0.8$ (decreasing from 2.15×10^5 to 2.11×10^5 m³), while the carbon footprint shows even smaller variation of only 0.7 % (from 1.35×10^5 to 1.34×10^5 kg CO₂e). Notably, external water transfer demand is most sensitive to uncertainty. When λ increases from 0.5 to 0.9, external water transfer volume increases dramatically from 0.90×10^5 m³ to 1.44×10^5 m³, representing an increase of 60.6 %. This indicates that overly conservative strategies may lead to excessive dependence on external resources.

5. Discussion

5.1. Synergistic management of resources for seasonal variations and regional differences

This study constructs a dynamic multi-objective optimization model within a Water-Energy-Food-Carbon Nexus (WEFEN) framework, extending the traditional Water-Energy-Food (WEF) coupling system to include environmental impacts. We employ dynamic programming to capture seasonal variations, introduce fuzzy credibility constrained programming to address uncertainties in water resource allocation, and utilize the NSGA-II algorithm to solve multi-objective optimization problems, forming a comprehensive methodological framework for the MRB water resource allocation from which we obtained several key findings.

First, the optimization results reveal both the synergies and trade-offs between the various objectives within the WEFEN system. Economic benefits (F1) and clean energy production (F2) show a partial

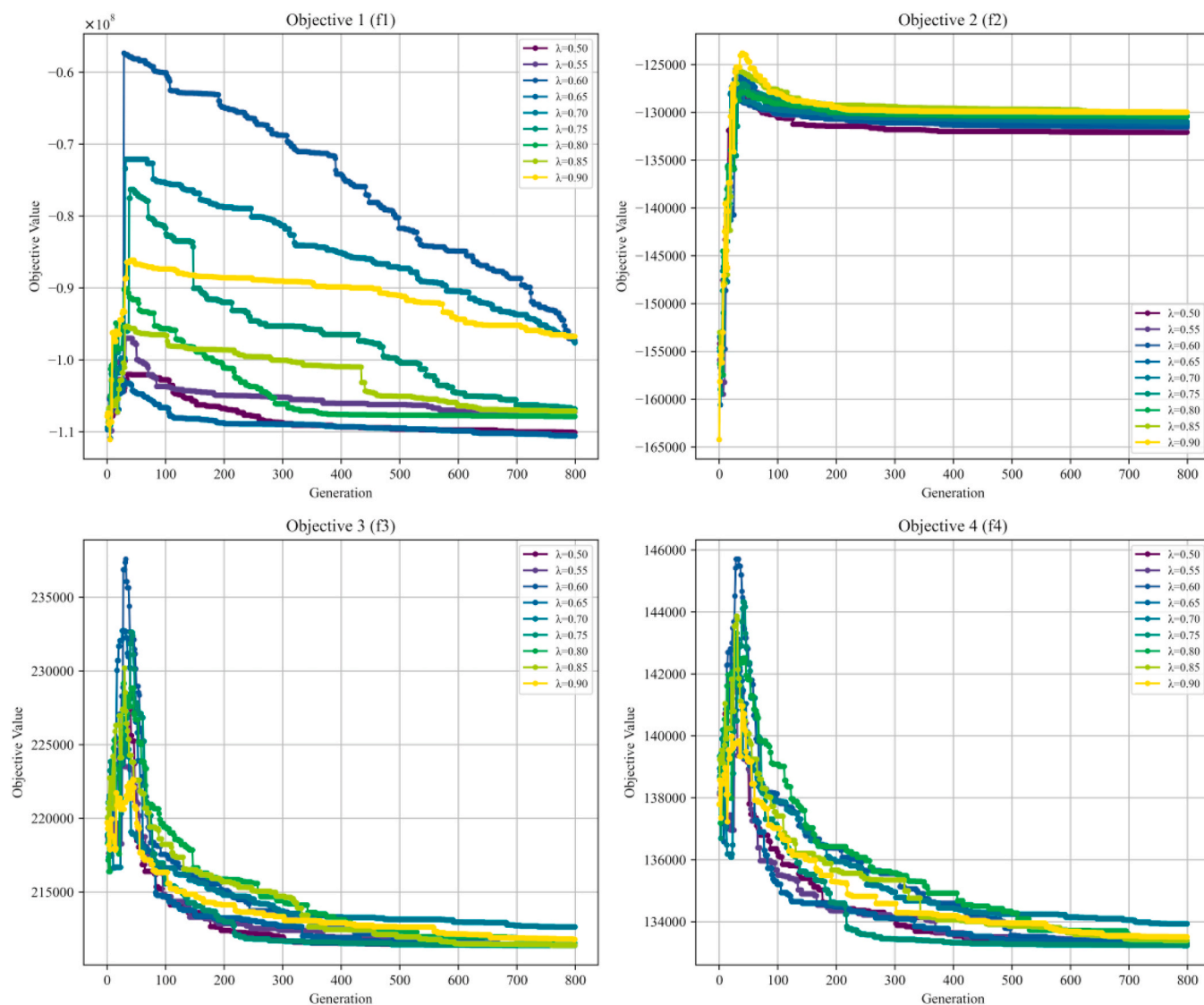


Fig. 14. Objective function values under different credibility levels.

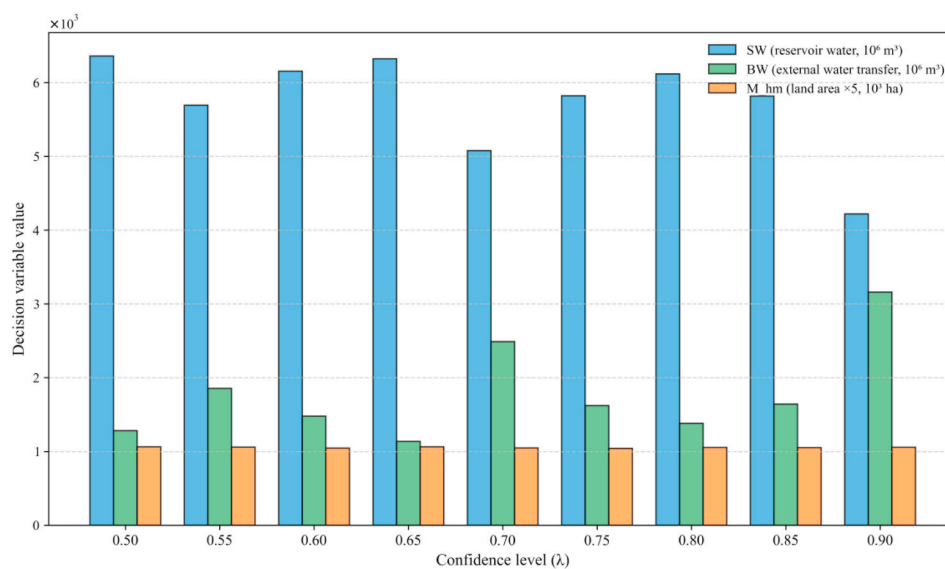


Fig. 15. Decision variable trends with changes in credibility level.

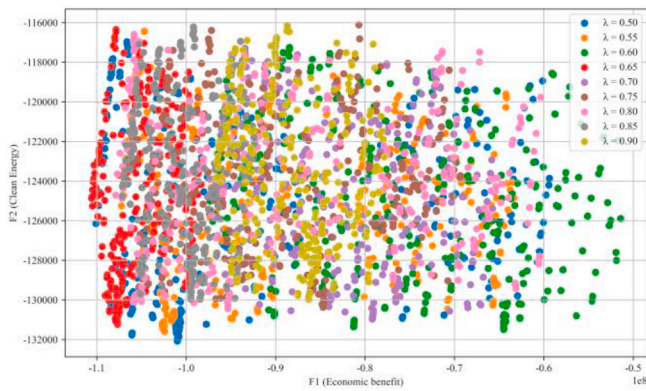


Fig. 16. F1 and F2 pareto frontier diagram.

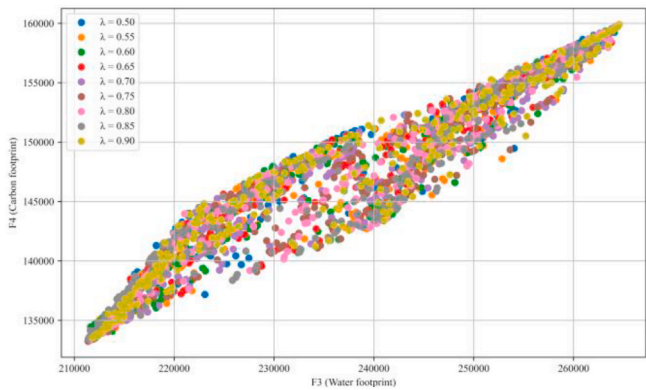


Fig. 17. F3 and F4 pareto frontier diagram.

positive correlation, suggesting that optimized water resource allocation can enhance both of these objectives simultaneously during spring, summer, and winter. Water footprint (F3) and carbon footprint (F4) exhibited strong linear positive correlation due to their association with planting areas. Clean energy output (F2) and water footprint (F3) displayed nonlinear negative correlation, reflecting the inherent conflict between energy production and water resource consumption. These links provide a theoretical and empirical foundation for designing policy for resource management strategies.

Second, we observe significant regional differences in water resource management. Upstream areas (such as Aba and Ya'an) have abundant water resources but lower energy conversion efficiency per unit of water resource (e.g., Aba at only 0.18 kWh/m^3) compared to the middle and lower reaches. Although midstream and downstream areas (such as Chengdu and Yibin) have higher economic benefits and food output, they face water resource shortages and greater environmental pressure. Chengdu and Yibin have the highest carbon emissions ($2.91 \times 10^4 \text{ kgCO}_2\text{e}$ and $2.73 \times 10^4 \text{ kgCO}_2\text{e}$ respectively), with grey water footprints peaking in autumn reflecting regional imbalances in resource-environmental pressure. Despite regional differences, our model manages to achieve efficient cross-regional resource allocation.

Third, this study makes several theoretical contributions. First, we expanded the traditional WEF nexus framework by constructing a WEFEN system optimization model that includes carbon and water footprints. While research traditionally focused on resource allocation efficiency and economic output, there was an overall lack of consideration of environmental impacts. This study incorporated carbon and water footprints as independent objective functions, achieving collaborative consideration of resource allocation and environmental impacts, thus developing a more comprehensive analysis of resource management. In the context of “dual carbon” goals, this framework provides a

theoretical foundation for coordinated research on agricultural water resource management and carbon emission reduction.

Fourth, the study combines dynamic programming with fuzzy credibility constrained programming, constructing a comprehensive optimization method for handling temporal and stochastic uncertainties (Pahl-Wostl, 2019). Compared to existing research that mostly employs static programming or handles only a single source of uncertainty, this study simultaneously captures seasonal changes and random disturbances affecting water resource systems. By parameterizing the uncertainty with credibility level λ , we quantitatively analyzed resource allocation strategies under uncertainty, addressing limitations of traditional deterministic models in complex uncertain environments.

5.2. Seasonal trade-offs between economic returns and carbon emissions

Higher economic returns can be realized in the spring and summer months, when carbon emissions are relatively low, mainly due to abundant water resources and more stable weather conditions. Clean energy production, especially hydropower and biomass, is higher during these two seasons, providing a stable supply of energy to support both agricultural production and economic activity in the non-agricultural sector. According to the figures in Table 7 below, clean energy production in spring and summer is $1.78 \times 10^5 \text{ kWh}$ and $3.15 \times 10^5 \text{ kWh}$, respectively, with carbon emissions of $1.34 \times 10^5 \text{ kg CO}_2\text{e}$ and $1.33 \times 10^5 \text{ kg CO}_2\text{e}$ respectively. This complementary nature of the resources allows for low carbon emissions and high economic returns to go hand in hand; autumn presents a different challenge.

In autumn, there is a significant drop in the productivity of hydropower due to water stress and increased competition for demand between agriculture and energy production. Also, continued high economic returns often require reliance on external, non-clean/renewable energy inputs, which inevitably increases carbon emissions. As a result, a clear negative trade-off between clean energy production and maximizing economic returns emerges in autumn. The need to expand carbon emissions while sacrificing low-carbon goals while maintaining economic growth reveals the “tension” faced in water and energy use in autumn, especially when resource conditions are limited, and balancing environmental protection and economic development becomes a more complex decision-making problem.

5.3. Uncertainty management discussion

The uncertainty analysis results provide important guidance for WEFEN system management. The study finds that the $\lambda = 0.70\text{--}0.80$ interval achieves the optimal balance between risk management and economic efficiency. Within this interval, economic benefits are maintained at $1.05\text{--}1.08 \times 10^8 \text{ CNY}$ while environmental pressures fall within a reasonable range. Policymakers should carefully consider this range of uncertainty which properly balances availability of resources with efficiency.

Seasonal analysis reveals that although autumn achieves the highest economic benefits ($12.43 \times 10^8 \text{ CNY}$), it also faces the greatest resource pressure, with external water transfers in Leshan and Meishan reaching 3.25×10^5 and $3.30 \times 10^5 \text{ m}^3$ respectively. In comparison, spring and winter show lower economic benefits but more stable system operations.

Table 7
Optimization results by season.

Seasons	F1: Economic benefits (10^8 CNY)	F2: Clean energy production (10^5 kWh)	F3: Water footprint (10^5 m^3)	F4: Carbon emissions ($10^5 \text{ kg CO}_2\text{e}$)
Spring	1.42	1.78	2.28	1.34
Summer	2.66	3.15	0.95	1.33
Autumn	12.43	5.88	1.45	1.37
Winter	1.07	1.21	1.85	1.32

This suggests the need to establish seasonal resource reserve and demand regulation mechanisms.

Regional disparity analysis indicates that upstream regions (Aba) have lower clean energy conversion efficiency (0.18 kWh/m^3), while downstream regions (Yibin, Leshan) exhibit higher efficiency (0.6 kWh/m^3) but face greater water resource pressure. It is recommended to establish cross-regional coordination mechanisms to leverage the synergistic effects of upstream water conservation and downstream efficient utilization. When the credibility level exceeds 0.85, external water transfer costs increase sharply and economic benefits decline significantly, indicating the existence of an efficiency boundary for resilience building. Policy design should avoid efficiency losses caused by overly conservative approaches.

6. Conclusion and policy implications

This study develops a dynamic Water-Energy-Food-Ecosystem Nexus (WEFEN) optimization framework addressing sustainable agricultural development under water scarcity, climate change, and resource allocation uncertainty. Integrating fuzzy credibility-constrained programming with multi-objective optimization via NSGA-II, the model simultaneously maximizes economic benefits and clean energy production while minimizing water and carbon footprints. Applied to the MRB, the framework yields several findings: (1) optimized resource allocation increases agricultural income by 37.61 % while reducing water and carbon footprints by 17.24 % and 9.20 %, respectively; (2) credibility levels of $\lambda = 0.70$ – 0.80 achieve optimal balance between economic returns and environmental sustainability, though $\lambda > 0.8$ increases external water diversion costs by 6.8 %; (3) seasonal variations create substantial allocation challenges, with water demand fluctuating 60 % across seasons, clean energy output varying from $1.21 \times 10^5 \text{ kWh}$ (winter) to $5.88 \times 10^5 \text{ kWh}$ (autumn), and autumn's high productivity requiring 40 % higher external water diversion to maintain production; (4) regional disparities show upstream areas face lower environmental pressure due to limited agricultural production, while downstream areas experience higher water demand and carbon emissions but achieve greater energy conversion efficiency.

Our findings yield three actionable policy recommendations for sustainable basin management. First, given the seasonal fluctuation in water demand and autumn's higher external water diversion requirement, policymakers should implement seasonal adaptive resource allocation mechanisms. This includes establishing quarterly allocation committees with targeted infrastructure subsidies during spring and winter as preparation periods. Additionally, emergency coordination centers with rapid-response funds should be deployed for peak demand seasons to address acute resource shortages. Second, recognizing the substantial regional disparities where upstream areas experience lower environmental pressure while downstream regions face higher water demand and carbon emissions, basin authorities should establish cross-regional coordination frameworks. These authorities should create basin-level councils with statutory authority to implement ecological payment systems that compensate upstream conservation areas, funded through downstream industrial levies. Technology transfer centers should be deployed in high-consumption regions to improve water-use efficiency and reduce downstream environmental pressures. Third, responding to the finding that credibility levels of $\lambda = 0.70$ – 0.80 optimize economic-environmental balance while $\lambda > 0.8$ increases external water diversion costs by 6.8 %, water resource managers should deploy risk-calibrated resource management systems. These systems should incorporate real-time monitoring networks (e.g., 5 stations per 1000 km^2) with automated threshold protocols that trigger emergency reallocation when $\lambda < 0.70$ and initiate cost-optimization reviews when $\alpha > 0.80$. An early warning system should complement these mechanisms to proactively identify and manage emerging resource conflicts before escalation.

Future research could integrate climate change scenarios and

extreme weather (such as the 2022 heat wave and drought) events to enhance system resilience, develop boundary sensitivity analysis for optimal spatial WEFEN boundaries, incorporate real-time monitoring for adaptive management, test framework transferability across different regions, and explore market-based mechanisms including carbon pricing and water trading to strengthen policy implementation effectiveness.

CRedit authorship contribution statement

Mingkang Yuan: Writing – original draft, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Jiabei Hu:** Writing – original draft, Visualization, Resources, Methodology, Investigation, Data curation. **Gavin Kader:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Formal analysis, Conceptualization. **Peng Lei:** Writing – review & editing, Visualization, Supervision, Resources, Investigation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Mingkang Yuan reports financial support was provided by Philosophy and Social Science Research Fund of Chengdu University of Technology (No. YJ2025-CZ009), Yangtze River Key Ecological Functional Area Protection Policy Research Center (No. YREPC2024-YB003), Sichuan Mineral Resources Research Center (No. SCKCZY2024-ZD001), and Energy and Environment Carbon Neutrality Innovation Research Center (No. ZC04202403). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- Ali, S.M., Acquaye, A., 2024. An examination of water-energy-food nexus: from theory to application. *Renew. Sustain. Energy Rev.* 202, 114669.
- Bertsekas, D., 1982. Distributed dynamic programming. *IEEE Trans. Automat. Control* 27, 610–616.
- Bieber, N., Ker, J.H., Wang, X., et al., 2018. Sustainable planning of the energy-water-food nexus using decision making tools. *Energy Policy* 113, 584–607.
- Brahmi, M.A., Dahane, M., Souier, M., et al., 2022. Sustainable capacitated facility location/network design problem: a non-dominated sorting genetic algorithm based multiobjective approach. *Ann. Oper. Res.* 311, 821–852.
- Cao, W., Wang, F., Chen, K., et al., 2024. Potential of achieving sustainable development through integrated optimization of mariculture species based on the food-energy-water nexus in China. *Sustain. Prod. Consum.* 49, 208–218.
- Chen, J., Wu, H., Zhang, C., et al., 2023. An inexact bi-level multi-objective modeling approach for managing agricultural water-food-carbon-land-ecology-nutrition nexus in an arid area. *J. Hydrol.* 621, 129556.
- Conway, D., Van Garderen, E.A., Deryng, D., et al., 2015. Climate and southern Africa's water-energy-food nexus. *Nat. Clim. Change* 5, 837–846.
- Correa-Cano, M.E., Salmoral, G., Rey, D., et al., 2022. A novel modelling toolkit for unpacking the water-energy-food-environment (WEFE) nexus of agricultural development. *Renew. Sustain. Energy Rev.* 159, 112182.
- Crippa, M., Solazzo, E., Guizzardi, D., et al., 2021. Food systems are responsible for a third of global anthropogenic GHG emissions. *Nat. Food* 2, 198–209.
- Cui, S., Wang, S., 2024. Tracking water pressure on water-energy-food system in global trade. *Sustain. Prod. Consum.* 50, 227–238.
- Cui, S., Zhang, J., Wang, X., et al., 2023. Fuzzy composite risk assessment of water-energy-food-carbon nexus in the dispark pumped irrigation system. *J. Hydrol.* 624, 129879.
- Demir, E., Alp, E., 2025. A framework for assessing the circular economy potential in the water and agriculture sectors in Türkiye through the water-energy-food-ecosystem nexus. *Sustain. Prod. Consum.* 54, 335–347.
- Duan, W., Chen, Y., Zou, S., et al., 2019. Managing the water-climate-food nexus for sustainable development in Turkmenistan. *J. Clean. Prod.* 220, 212–224.
- Duan, W., Zou, S., Chen, Y., et al., 2020. Sustainable water management for cross-border resources: the balkhash Lake Basin of Central Asia, 1931–2015. *J. Clean. Prod.* 263, 121614.

- Elahi, E., Zhu, M., Khalid, Z., et al., 2024. An empirical analysis of carbon emission efficiency in food production across the Yangtze River basin: towards sustainable agricultural development and carbon neutrality. *Agric. Syst.* 218, 103994.
- Emamjomehzadeh, O., Omid, F., Kerachian, R., et al., 2024. Water-energy-food-greenhouse gases nexus management in urban environments: a robust multi-agent decision-support system. *Sustain. Cities Soc.* 113, 105676.
- Fan, J.L., Kong, L.S., Zhang, X., 2018. Synergetic effects of water and climate policy on energy-water nexus in China: a computable general equilibrium analysis. *Energy Policy* 123, 308–317.
- Fan, X., Peng, J., Han, Y., et al., 2025. Water-energy-food nexus in the sustainable management of crop-livestock coupled systems. *Appl. Energy* 378, 124824.
- Feng, T., Liu, B., Ren, H., et al., 2025. A resource bi-level optimization model of water-energy-food-carbon nexus in a regional agricultural system under multi-stakeholder conditions. *Renew. Energy* 241, 122321.
- Garcia, D.J., You, F., 2016. The water-energy-food nexus and process systems engineering: a new focus. *Comput. Chem. Eng.* 91, 49–67.
- Garau, E., de Abreu, G., Pérez-Ramírez, I., et al., 2025. Exploring agricultural stakeholders' mental models of the water-energy-food-ecosystems (WEFE) nexus: insights from Mediterranean case studies. *Environ. Sci. Pol.* 171, 104196.
- Guo, L., Wu, Y., Huang, F., et al., 2024. An approach to complex transboundary water management in central Asia: evolutionary cooperation in transboundary basins under the water-energy-food-ecosystem nexus. *J. Environ. Manag.* 351, 119940.
- He, L., Xu, Z., Wang, S., et al., 2022. Optimal crop planting pattern can be harmful to reach carbon neutrality: evidence from food-energy-water-carbon nexus perspective. *Appl. Energy* 308, 118364.
- Hua, E., Engel, B.A., Guan, J., et al., 2022. Synergy and competition of water in food-energy-water nexus: insights for sustainability. *Energy Convers. Manag.* 266, 115848.
- Hua, E., Han, X., Bai, Y., et al., 2023. Synergy of water use in water-energy-food nexus from a symbiosis perspective: a case study in China. *Energy* 283, 129164.
- Ju, K., Wang, J., Wei, X., et al., 2023. A comprehensive evaluation of the security of the water-energy-food systems in China. *Sustain. Prod. Consum.* 39, 145–161.
- Kertoli, E., Prosperi, P., Harbouze, R., et al., 2024. The water-energy-food-ecosystem nexus in North Africa dryland farming: a multi-criteria analysis of climate-resilient innovations in Morocco. *Agric. Food Econ.* 12, 34.
- Khanjani Shiraz, R., Tavana, M., Fukuyama, H., et al., 2017. Fuzzy chance-constrained geometric programming: the possibility, necessity and credibility approaches. *Oper. Res.* 17, 67–97.
- Khofi, L., Manderson, L., Reid, M., et al., 2025. A review of the social and local dynamics in South Africa's water-energy-food nexus. *J. Environ. Manag.* 380, 124938.
- Li, Y., Zhang, L., Zhang, P., et al., 2024. Water-energy-food nexus in China: an interregional comparison. *Agric. Water Manag.* 301, 108964.
- Liu, G., Du, S., Gao, Y., et al., 2024. A study on energy-water-food-carbon nexus in typical Chinese northern rural households. *Energy Policy* 188, 114100.
- Liu, H., Fang, Z., Li, R., 2022. Credibility-based chance-constrained multimode resource-constrained project scheduling problem under fuzzy uncertainty. *Comput. Ind. Eng.* 171, 108402.
- López, J.C.L., Noriega, J.J.S., Figueira, J.R., et al., 2021. Non-dominated sorting genetic-based algorithm for exploiting a large-sized fuzzy outranking relation. *Eur. J. Oper. Res.* 293, 615–631.
- Lucca, E., El Jitany, J., Castelli, G., et al., 2023. A review of water-energy-food-ecosystems Nexus research in the Mediterranean: evolution, gaps and applications. *Environ. Res. Lett.* 18, 083001.
- Ma, Y., Li, Y., Huang, G., et al., 2024. Collaborative management of water-energy-food-ecosystems nexus in Central Asia under uncertainty. *Water Resour. Res.* 60, e2023WR035166.
- Mansour, F., Al-Hindi, M., Abou Najm, M., et al., 2023. Multi-objective optimization for comprehensive water, energy, food nexus modeling. *Sustain. Prod. Consum.* 38, 295–311.
- Martinez-Hernandez, E., Leach, M., Yang, A., 2017. Understanding water-energy-food and ecosystem interactions using the nexus simulation tool NexSym. *Appl. Energy* 206, 1009–1021.
- Melo, F.P.L., Parry, L., Brancalion, P.H.S., et al., 2021. Adding forests to the water-energy-food nexus. *Nat. Sustain.* 4, 85–92.
- Napoli, A., Matiu, M., Laiti, L., et al., 2025. Review on climate change impacts on the water-energy-food-ecosystems (WEFE) Nexus in the North-Eastern Italian alps. *Clim. Change* 178, 41.
- Nika, C.E., Vasilaki, V., Renfrew, D., et al., 2022. Assessing circularity of multi-sectoral systems under the water-energy-food-ecosystems (WEFE) nexus. *Water Res.* 221, 118842.
- Omrani, H., Alizadeh, A., Emrouznejad, A., et al., 2021. Data envelopment analysis model with decision makers' preferences: a robust credibility approach. *Ann. Oper. Res.* 1–38.
- Ortiz-Bobea, A., Ault, T.R., Carrillo, C.M., et al., 2021. Anthropogenic climate change has slowed global agricultural productivity growth. *Nat. Clim. Change* 11, 306–312.
- Pahl-Wostl, C., 2019. Governance of the water-energy-food security nexus: a multi-level coordination challenge. *Environ. Sci. Pol.* 92, 356–367.
- Pan, M., Tang, Z., Zhao, G., 2025. Organic farming enhances the synergy of the water-energy-food-ecology nexus. *Agric. Syst.* 224, 104247.
- Panjehpour, A., Zarrinpoor, N., 2025. A network design of a canola-to-biodiesel supply chain by considering the water-energy-food-land-ecosystem-undesirable climate change nexus. *Comput. Ind. Eng.* 199, 110610.
- Peykani, P., Hosseinzadeh Lotfi, F., Sadjadi, S.J., et al., 2022. Fuzzy chance-constrained data envelopment analysis: a structured literature review, current trends, and future directions. *Fuzzy Optim. Decis. Making* 1–65.
- Qin, J., Duan, W., Chen, Y., et al., 2022. Comprehensive evaluation and sustainable development of water-energy-food-ecology systems in Central Asia. *Renew. Sustain. Energy Rev.* 157, 112061.
- Qin, J., Duan, W., Zou, S., et al., 2024. Global energy use and carbon emissions from irrigated agriculture. *Nat. Commun.* 15, 3084.
- Rasul, G., Sharma, B., 2016. The nexus approach to water-energy-food security: an option for adaptation to climate change. *Clim. Policy* 16, 682–702.
- Ren, H., Liu, B., Zhang, Z., et al., 2022. A water-energy-food-carbon nexus optimization model for sustainable agricultural development in the Yellow River Basin under uncertainty. *Appl. Energy* 326, 120008.
- Sanchez-Zarco, X.G., Ponce-Ortega, J.M., 2023. Water-energy-food-ecosystem nexus: an optimization approach incorporating life cycle, security and sustainability assessment. *J. Clean. Prod.* 414, 137534.
- Shannak, S., 2022. Optimizing dynamics of water-energy-food nexus in a desert climate. *Energy Policy* 164, 112884.
- Sievers, E., Canovas, I., Kristensen, D., et al., 2025. Assessing to act: a water-energy-food-ecosystem (WEFE) nexus governance assessment for the Inkomati-Usuthu river basin in South Africa. *Environ. Sci. Pol.* 164, 103986.
- Springmann, M., Clark, M., Mason-D'Croz, D., et al., 2018. Options for keeping the food system within environmental limits. *Nature* 562, 519–525.
- Sušnik, J., 2018. Data-driven quantification of the global water-energy-food system. *Resour. Conserv. Recycl.* 133, 179–190.
- Sutar, M., Jadhav, H.T., 2023. A modified artificial bee colony algorithm based on a non-dominated sorting genetic approach for combined economic-emission load dispatch problem. *Appl. Soft Comput.* 144, 110433.
- Tan, Z., Li, H., Song, Q., et al., 2025. Synergistic optimization and interaction evaluation of water-energy-food-ecology nexus under uncertainty from the perspective of urban agglomeration. *Sustain. Cities Soc.* 106291.
- Vahedi, R., Ledari, M.B., Fani, M., 2025. Advancing circular economy in the water-energy-food nexus: a framework for sustainable resource use. *J. Environ. Manag.* 376, 124324.
- Wichelns, D., 2017. The water-energy-food nexus: is the increasing attention warranted, from either a research or policy perspective? *Environ. Sci. Pol.* 69, 113–123.
- Wu, H., Yue, Q., Guo, P., et al., 2025. Exploiting the potential of carbon emission reduction in cropping-livestock systems: managing water-energy-food nexus for sustainable development. *Appl. Energy* 377, 124443.
- Yang, G., Li, X., Huo, L., et al., 2020. A solving approach for fuzzy multi-objective linear fractional programming and application to an agricultural planting structure optimization problem. *Chaos Solitons Fractals* 141, 110352.
- Yang, G., Su, Y., Huo, L., et al., 2025. A multi-objective synergistic optimization model considering the water-energy-food-carbon nexus and bioenergy. *Agric. Water Manag.* 312, 109431.
- Yang, Y., Yang, L., Gao, H., 2025. A novel assessment framework for analyzing the sustainability of water-energy-food nexus system from a pressure-support perspective. *Energy Convers. Manag.* 332, 119715.
- Yoon, P.R., Lee, S.H., Choi, J.Y., et al., 2022. Analysis of climate change impact on resource intensity and carbon emissions in protected farming systems using water-energy-food-carbon Nexus. *Resour. Conserv. Recycl.* 184, 106394.
- Yuan, M., Chen, X., Ren, H., et al., 2023. Dynamic multi-period sustainable water resources optimal allocation strategies: a case study of China. *Comput. Ind. Eng.* 186, 109713.
- Yusuf, N., Baldacci, R., AlNouss, A., et al., 2024. Multi-period optimisation of flexible natural gas production network infrastructure with an operational perspective: a mixed integer linear programming approach. *Energy Convers. Manag.* X 24, 100818.
- Zhang, C., Wei, Y., Zhao, X., et al., 2025. Assessment and enhancement pathways of the water-energy-food-economy-ecosystem nexus in China's yellow river basin. *Energy* 316, 134492.
- Zhang, T., Tan, Q., Yu, X., et al., 2020. Synergy assessment and optimization for water-energy-food nexus: modeling and application. *Renew. Sustain. Energy Rev.* 134, 110059.
- Zhou, Y., Zhang, W., Kang, J., et al., 2021. A problem-specific non-dominated sorting genetic algorithm for supervised feature selection. *Inf. Sci.* 547, 841–859.
- Zolghadr-Asli, B., McIntyre, N., Djordjevic, S., et al., 2023. The sustainability of desalination as a remedy to the water crisis in the agriculture sector: an analysis from the climate-water-energy-food nexus perspective. *Agric. Water Manag.* 286, 108407.